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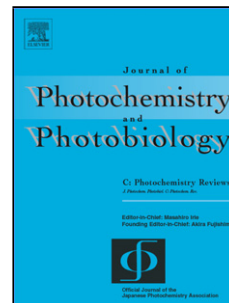
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Molecular design for efficient singlet fission

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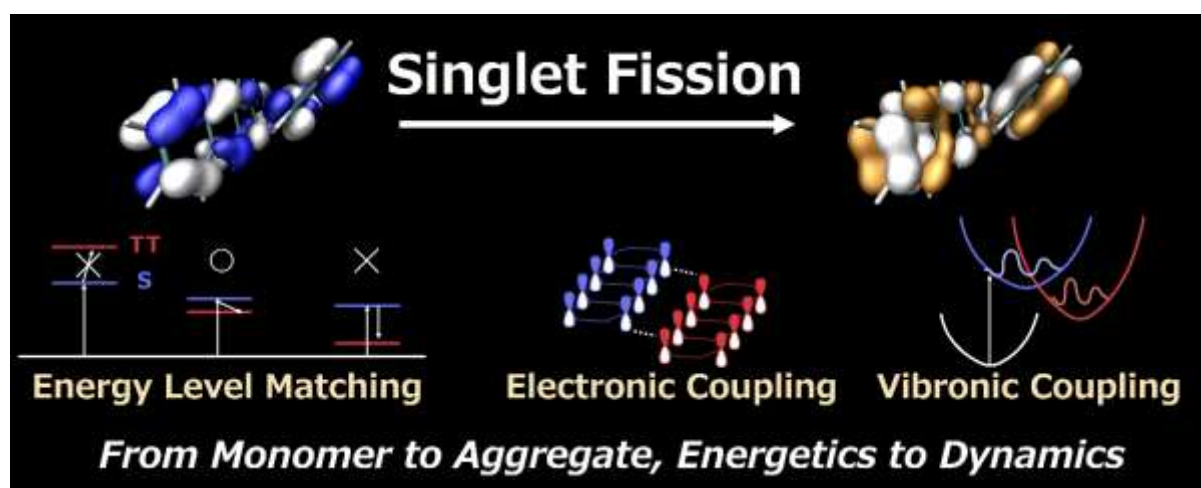
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Graphical Abstract



Highlights

- Theoretical and experimental analyses of mechanism and molecular design of efficient singlet fission.
- Singlet fission based on bottom-up theoretical modeling from a molecule to a molecular aggregate or crystal.
- Guidelines and strategies for designing efficient singlet fission.
- Realistic singlet fission systems and their performance and connections with molecular structures.

ABSTRACT

Singlet fission is a photophysical process in molecules and molecular aggregates, in which a singlet exciton generated by irradiation splits into two triplet excitons. Recently, singlet fission has attracted a great deal of attention from the viewpoint of applications in organic photovoltaic cells, where singlet fission has a possibility of improving the photoelectric conversion efficiency. Although singlet fission was first observed about 50 years ago in anthracene crystals, and the mechanism has been investigated in detail for a small number of molecular systems such as tetracene and pentacene crystals, the relationships between molecular or crystal structures and singlet fission efficiency are yet to be precisely clarified. Thus, molecular structure – singlet fission relationships and molecular or crystal design guidelines for efficient singlet fission are intensely desired for realizing efficient photovoltaic energy conversion, and experimental and theoretical investigations advance rapidly. We introduce three investigation steps, which are based on bottom-up theoretical modeling from a molecule to a molecular aggregate or crystal. The modelling involves energy level matching at the molecular level, electronic coupling at the aggregate level, and singlet fission dynamics including exciton–phonon (vibronic) coupling, by emphasizing the importance of interplay between each step. From the modelling, we present several design guidelines for efficient singlet fission, which is together with practical molecular structures, chemical modifications and molecular configurations.

Contents

1. Introduction
2. Singlet fission process
3. Energy level matching conditions
 - 3.1. Relationship between excitation energies and electronic states of a single molecule
 - 3.2. Diradical character
 - 3.3. Relationship between multiple diradical character and excitation energies for H₄ model
 - 3.4. Diradical character based molecular design guideline for singlet fission
 - 3.5. Diradical character based design of several singlet fission molecules
4. Electronic coupling
 - 4.1. Model Hamiltonian approach
 - 4.2. Singlet fission in pancake-bonded systems
 - 4.3. Intramolecular singlet fission in covalently-linked systems: singly-bridged systems
 - 4.4. Intramolecular singlet fission in covalently-linked systems: doubly-bridged systems
5. Exciton dynamics in singlet fission
 - 5.1. Vibronic coupling
 - 5.2. Vibronic coupling density analysis
 - 5.3. Theory of exciton dynamics, including exciton-phonon coupling
 - 5.4. Applications of exciton dynamics, including exciton-phonon coupling
 - 5.5. State-dependent spectral density effects on singlet fission dynamics
6. Summary and perspectives

1. Introduction

Singlet fission (SF), or singlet exciton fission, is a photophysical phenomenon, in which a singlet exciton splits into two triplet excitons having long lifetimes in crystal/solution phase of organic molecular aggregates or in molecules. Singlet fission was a mechanism originally proposed in the 1960s in order to interpret the significant reduction of fluorescence yields observed for anthracene and tetracene crystals [1,2]. By the mid-1970s, many studies have been conducted on the elucidation of the mechanism including magnetic field effects on the singlet fission [3] as well as on the reverse processes, i.e., triplet–triplet annihilation (TTA) [4], where two triplet excitons collide and fuse into a singlet exciton. The results on singlet fission studies at the time have been summarized in reviews [5,6]. Although the singlet fission was also observed in conjugated polymer and carotenoids rather than acenes in the 1980s [7], the interest in the singlet fission had gradually dropped off. However, it has started receiving attention again since Nozik and co-workers have emphasized its importance in the application to photovoltaics in 2004 [8,9], where they proposed a possibility of improving photoelectric conversion efficiency from the Shockley–Queisser limit ~33% to ~47% through multi-exciton generation due to singlet fission. Subsequently, a series of theoretical and experimental studies by Michl and co-workers suggested a promising potential of well-designed materials for an application in singlet fission [10–12].

Many works have been devoted to investigating photovoltaic cells application of singlet fission. Triplet excitons from singlet fission can be generated with ideal 200% yields (two triplet excitons per a singlet exciton) in several chromophore aggregates [12–16]. Indeed, external quantum efficiency beyond 100% by singlet fission was observed [17]. Power conversion efficiency using singlet fission cells were also examined [18–22], and was reported up to 4.8% [20]. More fundamental understanding in singlet fission mechanisms and constructing efficient SF materials design guidelines are intensely desired to boost up this situation.

There have been lots of investigations on the detailed mechanism of singlet fission, design for highly efficient singlet fission molecules and aggregates, together with measurements of the singlet fission of novel molecules and crystals. Recently, several experimental and theoretical investigations have been conducted particularly by focusing on the relationships between electronic structures, electronic couplings (intermolecular interactions), exciton-phonon (vibronic) couplings and singlet fission dynamics [23–33]. These results would contribute to elucidating the comprehensive mechanisms of the singlet fission as well as to building novel design guidelines for efficient singlet fission molecular systems, while the fact that there are still a small number of efficient singlet fission molecules/aggregates prevents further progress in the singlet fission study [23,24].

There already exist many reviews on singlet fission. Recent reviews are especially focusing on comprehensive overview of singlet fission, theoretical models, and several design guidelines [23], recent development of the research area [24,34], photophysics of a specific singlet fission chromophore [35–37], delocalization of excited states [38], *ab initio* theoretical methods for crystalline tetracene and pentacene [39], and energy conversion into electric current from singlet fission [21,37,40]. In this Review, therefore, we overview and summarize the recent studies on singlet fission, especially from the viewpoint of revealing the microscopic mechanism ranging from a single molecule to molecular aggregates, and also from energetics to dynamics. Furthermore, we particularly focus on how one finds design guidelines and construct design strategies for efficient singlet fission molecular systems. The guidelines are then applied to several realistic systems as examples to illuminate their performance as well as their close connections with molecular structures.

2. Singlet fission process

The singlet fission process is approximately described by [23]



Here, we consider two molecules in a crystal, and assume that one is in the first singlet excited state (S_1), while the other is in the ground state (S_0). T_1 is the lowest triplet state, and ${}^1(T_1T_1)$ (also referred to as TT or ${}^1(TT)$, hereafter) indicates the singlet state composed of two triplet excitons on adjacent molecules, referred to as “correlated triplet pair” or “double triplet”. Due to this singlet nature of correlated triplet pair, singlet fission is considered as spin-conserving process. Therefore, spin–orbit interaction is not required for singlet fission, which is distinguished from traditional triplet generation by intersystem crossing. The singlet fission occurs due to intermolecular interaction $H_{\text{int}} = H_{\text{el}} + H_{\text{sp}}$, where H_{el} indicates the spin-free electronic interaction Hamiltonian, and H_{sp} does the spin interaction Hamiltonian including Zeeman interaction with an external magnetic field and spin dipole–dipole interaction [23]. The first step of Eq. (1) described by rate constants $k_{-2/2}$ and the second one by $k_{1/-1}$ are artificially divided by separating actions of H_{el} and H_{sp} , respectively. Namely, the first step represents the internal conversion process, while the second one does the intersystem crossing process. It is noted again that this separation into two steps is artificial and, in reality, these processes proceed simultaneously. In principle, three spin adapted states, singlet ${}^1(T_1T_1)$, triplet ${}^3(T_1T_1)$, and quintet ${}^5(T_1T_1)$, are generated from the correlated two triplets (see Fig. 1). The relative energies between these states are primarily determined by H_{el} and affect the singlet fission efficiency. In the case of these states being near-degenerate, they are mixed by H_{sp} to create superposition states, which evolve and relax to two free triplets when the correlated triplets diffuse apart rapidly. Although, in principle, singlet fission process can create excited quintet states instead of triplets, such event was hardly observed due to the high energy levels of the excited quintet states [23]. Despite that, a very recent experimental work has succeeded in observing a quintet state generated from singlet fission [41]. On the other hand, a reverse process possibly exists, where the superposition states return to the original singlet excited state by triplet fusion or

triplet–triplet annihilation (TTA) [5], though at present a mechanism and design guidelines of controlling these two processes have not been elucidated sufficiently. Thus, the investigation of singlet fission process would contribute to the clarification of TTA as well as to controlling singlet – triplet exciton interconversion.

In this Review, we consider three substantial steps as a design strategy for efficient singlet fission materials. First, there exist several conditions on the monomer energy level since the first singlet excited state energy of the dimer is expected to be similar to that of the monomer in usual molecular crystal. This leads to some conditions on the energy levels of the monomer for singlet fission, which is referred to as “energy level matching conditions” [10,23] as explained in the next section. Second, since singlet fission is a phenomenon mostly occurring in molecular aggregates, the transition probability from S_1S_0 to $^1(TT)$ due to the intermolecular interaction (electronic coupling H_{el}) has to be evaluated. Even if the energy level matching conditions were satisfied, the small transition probability would reduce the singlet fission efficiency. Thus, the investigation of the electronic coupling aims at clarifying the relationships between the molecular configuration (molecular packing) and electronic couplings. Third, furthermore, molecular vibrations are also predicted to affect the singlet fission efficiency because the electron-phonon coupling (vibronic coupling) is the origin of the exciton relaxation (exciton migration). Thus, to reveal the whole picture of singlet fission mechanism and to construct the design guidelines for efficient singlet fission, we need to perform singlet fission dynamics simulation considering the molecular geometric and electronic structures (energy level matching), molecular configurations (electronic coupling), and relaxation factors based on the vibronic couplings. In summary, the above three steps of investigations for singlet fission materials are shown in Fig. 2, at each step of which we could clarify the mechanism and present design guidelines for improving singlet fission efficiency. Although experimental and theoretical investigations at each step progress rapidly, relationships between singlet fission efficiency and molecular/aggregate structures still remain unveiled, and correlations between the results obtained at each step on singlet fission process have hardly been clarified. For

more complete understanding of singlet fission mechanism as well as for constructing comprehensive singlet fission material design guidelines, further progress in investigations at each step as well as organization of the results obtained at each step have to be driven intensely through close cooperation between experimentalists and theorists in the related fields.

3. Energy level matching conditions

3.1. Relationship between excitation energies and electronic structures of a single molecule

From the viewpoint of energy level matching in singlet fission process, Michl et al. proposed the following conditions of excitation energies of a single molecule for singlet fission [23,24]:

$$2E(T_1) \sim E(S_1) \text{ or } 2E(T_1) < E(S_1), \quad (2)$$

$$2E(T_1) < E(T_2). \quad (3)$$

Eq. (2) indicates the condition required for split of a singlet exciton (with energy $E(S_1)$) to two triplet excitons (with energy $E(T_1)$), while Eq. (3) is required to suppress the recombination of the split two triplet excitons into another triplet exciton (with energy $E(T_2)$) represented by $T_1 + T_1 \rightarrow T_2 + S_0$.

From the viewpoint of decreasing the energy loss in the photoelectric conversion process, molecules satisfying $2E(T_1) \sim E(S_1)$ are expected to be appropriate for efficient singlet fission. When singlet fission is exothermic, $2E(T_1) - E(S_1) < 0$, the process is expected to undergo efficiently from the energetic viewpoint. For endothermic singlet fission systems, $2E(T_1) - E(S_1) > 0$, there may be an energy barrier to be overcome by thermal excitation, though the temperature effect is complicated and not fully understood yet [42–48]. We do not go in detail of temperature effects because we here focus on molecular design for efficient singlet fission, where we need to design energetically favorable (isothermic or exothermic) molecules for singlet fission. However, many thermally stable closed-shell molecules do not tend to satisfy the energy level matching condition because they have large $E(T_1)$ values near to $E(S_1)$ [23]. Michl et al. proposed two types of molecular species – “alternant hydrocarbon” and “biradicaloid” (diradical molecule) – as potential candidates for singlet

fission [23]. This is based on the consideration of exchange interactions between frontier orbitals in a molecule. On the other hand, there are also many exceptions, e.g., naphthalene and phenacene, which are not known to exhibit singlet fission. Diradical molecules are known to have small S_0-T_1 gap, which leads to relatively lower T_1 state than S_1 state so as to satisfy the first condition (Eq. (2)). However, the systems with too small S_0-T_1 gaps and those with triplet ground states are not suitable for efficient singlet fission because they have large deviation from $2E(T_1) \sim E(S_1)$, leading to high energy loss and slow singlet fission rate.

As discussed above, although design guidelines presented so far are useful to roughly determine the searching area and to find similar systems to existing singlet fission molecules, they are still insufficient to quest for unknown singlet fission molecules [49,50]. In order to improve these deficiencies as well as to obtain more reliable design guidelines for efficient singlet fission molecules, we have constructed a new design principle based on the diradical character [49]. The diradical character (γ) is quantum-chemically well defined by twice the weight of doubly excited configuration in the singlet ground state multi-configuration self-consistent-field (MC-SCF) wavefunction [51,52]. This index indicates the “bond weakness” or “electron correlation” of the singlet ground state molecule [51–54]. Furthermore, the definition of the diradical character can be extended to the description of multi-radical systems by using multiple diradical characters [52–57]. As shown in the next section, the energy level matching conditions are described quantitatively by the diradical character [49]. Since the diradical character is closely connected to the conventional chemical indices, e.g., bond order and aromaticity [53,54,58], such re-description of the energy level matching conditions is expected to contribute to developing the “chemist-friendly” design guidelines for efficient singlet fission based on practical chemical structures and modifications.

3.2. Diradical character

Before explaining diradical character based design guideline for singlet fission, we provide a brief outline of diradical character [51–57]. As mentioned in the previous section, although the

diradical character y_i (related to the HOMO- i and LUMO+ i) is originally defined by twice the weight of the doubly excited configuration in the MCSCF theory, we can use the broken symmetry approach like the spin-unrestricted Hartree-Fock (UHF) method to approximately evaluate the diradical characters [51,52]. Using the spin-projected UHF (PUHF) theory, which approximately removes the spin contaminations in the UHF solution, the diradical character y_i is formally expressed as [51,52]

$$y_i = 1 - \frac{2T_i}{1+T_i^2}, \quad (4)$$

where T_i , the orbital overlap between the corresponding orbital pairs. The diradical character takes a value between 0 and 1, which correspond to closed-shell and pure diradical states, respectively. The multiple diradical characters y_i ($i=0, 1, \dots$) indicate the multiradical character, where the number of radicals are estimated by twice the summation of all y_i values ($y_0 \geq y_1 \geq \dots$). The spatial distribution of these unpaired electrons is visualized by “odd electron” density [55,56,59], which is useful for understanding the chemical origin of the diradical character.

3.3. Relationship between multiple diradical character and excitation energies for H_4 model

Firstly, we introduce a classification of singlet fission molecules, which are classified into three classes I, II and III based on excitation configurations of the first singlet excited state, S_1 [23]. It is predicted that appropriate molecular orientations in crystal for singlet fission depends on these classes [23,24]. In class I, a primary configuration in S_1 state is the HOMO $\rightarrow$ LUMO singly excited configuration. It is known that most singlet fission molecules are classified into this class, e.g., tetracene, pentacene, and 1,3-diphenylisobenzofuran. Class II is characterized by the HOMO-1 $\rightarrow$ LUMO or HOMO $\rightarrow$ LUMO+1 singly excited configurations in the S_1 state. Several molecules with small zwitterionic resonance structure [60,61] and polycyclic hydrocarbons with $4n\pi$ electrons [62] are theoretically predicted to be classified into this class. In class III, the S_1 state is mainly described by doubly excited configurations including HOMO $\rightarrow$ LUMO double excitation. Note here that the low-lying singlet doubly excited state of polyenes and related linearly conjugated polymers is often

interpreted as a tetraradical excited state composed of two triplet excitons [63–65]. Several carotenoids composed of long polyene units, e.g., zeaxanthin [66], belong to this class, though its role in singlet fission is under discussion [67].

In order to clarify the relationships between diradical character and singlet fission efficiency, we consider a minimal model for the energy level matching conditions as well as for the above classification of singlet fission molecules. As shown in Eqs. (2) and (3), relative relationships between three excitation energies of $E(S_1)$, $E(T_1)$, and $E(T_2)$ are important for evaluating the energy level matching conditions for singlet fission. Considering a minimum model, a four-electron system can describe classes I, II and III, and also T_2 . As the simplest example of a four-electron system, we investigate a symmetric linear H_4 model (Fig. 4) [49], in which the singlet open-shell character (diradical (y_0) and tetraradical (y_1) characters) can be controlled by changing the interatomic distances, R_1 and R_2 [38]. The R_1 and R_2 are varied from 0.5 to 4.0 Å to span a whole range of multiple diradical characters (Fig. 4). In the H_4 model, the ground state is the lowest singlet state with A_g symmetry ($S_0 = 1^1A_g$), while T_1 and T_2 states are uniquely identified by the 1^3B_u and 1^3A_g states, respectively. On the other hand, there are two candidates for the first singlet excited state S_1 , i.e., 1^1B_u or 2^1A_g , where 1^1B_u is a singlet state including the HOMO $\rightarrow$ LUMO singly excited configuration, and 2^1A_g is a singlet state including the HOMO–1 (HOMO) $\rightarrow$ LUMO (LUMO+1) singly excited configuration and several doubly excited configurations. According to the classification of singlet fission (Fig. 4), $S_1 = 1^1B_u$ belongs to class I, and $S_1 = 2^1A_g$ to class II and/or III in the present model. For simplicity, class III is redefined here to include class II, i.e., the case of $S_1 = 2^1A_g$ belongs to class III.

As seen from Fig. 5d, $E(1^3B_u)$ decreases with increasing y_0 , and becomes 0 at the limit of $y_0 \sim 1$, while the other three excitation energies are relatively insensitive to y_0 and take similar stationary values except for their rapid decreases for $y_0 < \sim 0.2$. The diradical excitation energy $E(1^3B_u)$ is found to strongly correlate to diradical character in the ground state (y_0). Similarly, as shown in Fig. 4e,

$E(2^1A_g)$ and $E(1^3A_g)$ decrease with increasing y_1 , and finally become zero at the limit of $y_1 \sim 1$ because these behaviors originate in tetraradical characters of 1^1A_g , 2^1A_g and 1^3A_g states at $y \sim 1$ [49].

Therefore, the tetraradical excitation energies, $E(2^1A_g)$ and $E(1^3A_g)$, are found to strongly correlate to tetraradical character in the ground state (y_1). In contrast to the $E(1^3B_u)$, $E(2^1A_g)$ and $E(1^3A_g)$, $E(1^1B_u)$ is shown to be insensitive to the multiple diradical characters (y_0 and y_1), and has non-zero excitation energies in the whole region because of its pure ionic nature (Fig. 4d, e). As explained above, the important point is that the y_0 and y_1 dependences of excitation energies originate in the difference in characters of the wavefunctions between the ground and each excited state.

3.4. Diradical character based molecular design guideline for singlet fission

From the above results, the region satisfying the two energy level matching conditions is illustrated on the y_0 - y_1 plane together with the density plot of $\eta = [2E(1^3B_{1u}) - E(1^1B_{1u})]/E(1^1B_{1u})$ (see Fig. 5) [49], where the darker density exhibits the region with higher energy efficiency (~ 0 for the highest efficiency) of singlet fission from the absorption of light by the 1^1B_{1u} state, which is the lowest optically allowed state. The auxiliary lines, $[E(2^1A_g) - E(1^1B_{1u})] / E(1^1B_{1u}) = 0$ (solid) and -0.3 (dashed), are also shown to clarify the class of molecules for singlet fission. It is found that the energy level matching conditions are not satisfied in the closed-shell ($y_0 < 0.1$ in the present case) or in the tetraradical-like regions ($y_0 \sim y_1$). In addition, the energy efficiency of singlet fission is found to become worse with increasing y_0 due to the excess decrease in the lowest triplet excitation energy, $E(1^3B_u)$. It is also noted that the doubly excited state (2^1A_g) tends to be the lowest singlet excited state rather than the singly excited state (1^1B_{1u}) in the high y_1 region. This implies that class III is characterized by the more significant tetraradical nature than class I. In order to assess the reliability of this map, we examine the multiple diradical characters for tetracene (**A1**), 1,3-diphenylisobenzofuran (**A2**), pentacene (**A3**), and zeaxanthin (**B1**) (Fig. 5) [49], in which efficient singlet fissions are observed and group “A” and “B” belong to class I and III, respectively. These molecules are indeed found to lie in the open-shell region with $y_0 > 0.2$. In addition, group B exhibits

higher y_1 value than group **A**, where the S_1 state of the former is characterized by the doubly excited configurations and that of the latter is done by the HOMO $\rightarrow$ LUMO singly excited configuration, though group **A** lies slightly outside the region with $E(2^1A_g) - E(1^1B_{1u}) > 0$ in the present model. Nevertheless, the relative features between class I (**A1–3**) and III (**B1**) are distinguished on the y_0 – y_1 plane. These results indicate that the multiple diradical characters are useful for the semi-quantitative prediction of the feasibility of singlet fission though it is still difficult to predict the precise thresholds of multiple diradical characters and this condition is limited to the energy level matching for singlet fission. It is also found that the three polycyclic aromatic hydrocarbons **C1–C3** lie in the feasibility region of singlet fission. This theoretical prediction has been confirmed by directly checking the energy matching conditions [68] and also have recently verified experimentally using crystals of terrylene derivatives [14]. Using similar π -conjugation backbone, terrylenediimide derivatives were also shown to exhibit singlet fission [69,15]. A π -extended zethrene, heptazethrene, was suggested to possibly undergo singlet fission from the observation of a fast generation of long-lived species [70].

3.5. Diradical character based design of several singlet fission molecules

As an example of the application of the diradical character based design principle, we examine condensed-ring π -conjugated molecules with $4n\pi$ electrons ($n = 4, 5, \dots$), i.e., antiaromatic polycyclic hydrocarbons composed of five- and six-membered rings (see Fig. 6). These molecules are expected to exhibit non-negligible diradical character from their resonance structures and Clar's sextets rule, i.e., a larger number of Clar's sextets in the open-shell form than in the closed-shell form in a molecule leads to its intermediate diradical character. Judging from the excitation energies calculated using the TDDFT method, the energy level matching conditions are found to be satisfied in relatively small-size antiaromatic condensed-ring molecules with $y_0 \sim 0.4$ and $y_1/y_0 < 0.2$, i.e., intermediate diradical character without significant tetraradical character (indeno[1,2,3-*cd*]fluorene-like molecules **2(2)** and **2(3)**). A recent experimental study showed singlet fission in a derivative of **1(1)**,

though it is not from the lowest excited state (S_1) but from the second lowest excited state (S_2) [71]. Further investigations are desired for understanding the relationship between antiaromatic molecules and singlet fission.

As another example, we consider several heteroacene models involving nitrogen (N) atoms (see Fig. 7) [61]. The diradical character criterion is found to be closely related to the relative contributions of diradical and zwitterionic resonance structures of the heteroacenes, i.e., the aromaticity of the central ring(s). It turns out that the increase in the aromaticity of the central ring(s) prefers the diradical form to the zwitterionic form. From the comparison of the calculated excitation energies, it is found that two promising candidates (**N0-3** and **N2-4**), chosen based on the diradical character criterion, satisfy the energy level matching conditions. These two candidates are also found to belong to Class II and I, respectively. From the detailed analysis, the exchange integrals between frontier orbitals and orbital energy differences are found to play important roles for the classification of singlet fission molecules. It will be interesting to investigate the difference in singlet fission dynamics due to the different nature of S_1 states.

As the last example of diradical character based design of singlet fission molecules, we consider bisanthene (**C3**, or **BA1** in the following) and its chemical modifications [72]. From the excitation energy calculation, we find that the conditions Eq. (2) and (3) are satisfied together with somewhat large energy loss of $|2E(T_1) - E(S_1)| = 0.516$ eV [72]. Thus, we should reduce the diradical character y_0 by several chemical modifications in order to reduce this energy loss upon singlet fission. First, we consider attaching aromatic rings (**BA2-X**, Fig. 8) because this modification increases the number of sextets from two in **BA1**'s closed-shell form to four in **BA2-X**'s closed-shell form with keeping the number of sextets in the open-shell form of **BA2-X**. Due to increased sextets in the closed-shell form, their diradical character are found to be decreased relative to **BA1** (Table 1). Increased closed-shell nature of **BA2-X** can be visually examined by the resonance structures (Fig. 8) and by nucleus independent chemical shift on the π -plane [73] (Fig. 9). Next, we consider twisting

the π -plane by bulky groups, **BA3-X**. Because the HOMO and LUMO of **BA1** are composed of linear combinations of those of anthracene moieties in the molecule, π -plane twisting reduces the overlap between the HOMO and LUMO of parent anthracene moieties. Therefore, it is predicted that π -plane twisting increases the HOMO–LUMO gap and decreases the diradical character y_0 of **BA3-X**. Finally, we have obtained small energy loss chromophores **BA2-O** and **BA3-F**.

On the basis of the diradical character based design principle, several practical strategies for designing singlet fission molecules have been proposed [53,58,60–62,68,72,74–77]. For example, the use of captodative effect successfully introduced diradical character into the small π -systems [75]. It has been found that well-designed BN-substitution in naphthalenes and azulenes induce some amount of diradical character in them and they satisfy the energy level matching conditions [74]. Substitution effects through the introduction of N atoms and thiophene rings into oligoacenes have been investigated [78]. Recent study using the diradical-character-based screening approach successfully picked out 10 potential molecules from over 200 potential diradical compounds [77]. As another π -framework, it has been found that a quinoidal bithiophene is a very promising diradical system for intramolecular singlet fission with an outstanding efficiency of $\sim 180\%$ [79–81]. On the other hand, some experiments have shown that, even when singlet fission is energetically possible ($2E(T_1) < E(S_1)$), a large energy difference can suppress singlet fission. Some chemical modifications on pentacene by heteroatoms change the energy levels of pentacene more exoergic, and this has been found to cause less efficient singlet fission [82]. Increasing the π -conjugation length of pentacene to hexacene has been also examined. Again, too large thermodynamic force due to longer π -conjugation, that is, large diradical character [83], resulted in slower singlet fission than pentacene [84–86]. These strongly indicate the importance of “matching” in the energy levels for efficient singlet fission, that is, tuning of the diradical character.

The diradical character based design using singlet fission feasibility map (Fig. 5) can present a simple and an easy way of prescreening singlet fission candidates from a huge number of molecules

because the excitation energies concerning energy level matching conditions are usually difficult to be evaluated both experimentally and theoretically, while the diradical character in the ground state can be easily calculated, e.g., using spin-projected UHF calculations [29, 30]. Furthermore, the diradical character in the ground state is found to be closely related to various traditional chemical concepts and indices, e.g., effective bond order, aromaticity, and resonance structures [53,54,58]. Indeed, the two types of candidate systems proposed by Michl et al. [23,24], i.e., alternant hydrocarbons and biradicaloids, are shown to be understood in a unified way by a simple guideline based the multiple diradical characters, i.e., systems with weak diradical character and much smaller tetraradical character tend to be promising efficient singlet fission candidates, and the exception to the previous guideline can also be removed by the diradical character based guideline [49]. In summary, neither all the alternant hydrocarbons nor all the diradical molecules are always singlet fission candidates, but systems with relatively weak diradical characters and much weaker tetraradical characters become candidate molecules for efficient singlet fission from the energy level matching conditions.

4. Electronic coupling

The intermolecular interaction (referred to as electronic coupling) is predicted to have a strong impact on the singlet fission efficiency as mentioned in Sec. 2, in addition to the energy matching condition. There are two types of such interactions: non-covalent interactions in a crystal or an aggregate [48,87–102], and covalent interactions in a molecule composed of plural monomers [30,103–111,16,112–116]. For realizing efficient singlet fission, there are requirements for the interaction intensity as follows. In the case of competing with fluorescence, singlet fission has to occur in the pico- to subpico second time scale for its efficiency, the feature of which excludes too weak interactions. On the other hand, too strong interaction has to be avoided in order to facilitate the split of correlated triplet pair into free triplets, which requires near-degeneracy between $^1(\text{TT})$, $^3(\text{TT})$ and $^5(\text{TT})$. For example, too strong intermolecular interaction has a possibility of creating more

stabilized charge transfer state or excimer, which tends to open new relaxation pathways to the ground state S_0 and thus to reduce the singlet fission yield [23]. In this section, firstly, we explain how to determine the molecular configuration suitable for singlet fission based on the analysis of electronic coupling for dimer models with various molecular configurations [23,24]. Second, a new class of singlet fission aggregates, open-shell aggregates composed of π - π stacked phenalenyl radicals, are theoretically proposed. Finally, we cover a recent topic – a new class of intramolecular singlet fission in covalently-linked systems – and clarify its SF mechanism by using partitioning the electronic couplings into the chromophores and bridges. On the basis of this result, we present design guidelines for efficient singlet fission in chromophore dimers linked with π -conjugated bridges.

4.1. Model Hamiltonian approach

Let us consider a dimer composed of molecules, or chromophores, A and B. Michl et al. proposed the model Hamiltonian for singlet fission described by the six electronic states, that is, the ground state, singlet excited states ($S_1^A S_0^B$ and $S_0^A S_1^B$), charge-transfer states ($^1(C^A A^B)$ and $^1(A^A C^B)$), and the correlated triplet-pair state $^1(TT)$. All of them are totally spin singlet. The electronic Hamiltonian in these bases (excluding the ground state) is

$$H_{el} = \begin{pmatrix} \langle S_1 S_0 | H_{el} | S_1 S_0 \rangle & & & & & \\ \langle S_0 S_1 | H_{el} | S_1 S_0 \rangle & \langle S_0 S_1 | H_{el} | S_0 S_1 \rangle & & & & \\ \langle CA | H_{el} | S_1 S_0 \rangle & \langle CA | H_{el} | S_0 S_1 \rangle & \langle CA | H_{el} | CA \rangle & & & \\ \langle AC | H_{el} | S_1 S_0 \rangle & \langle AC | H_{el} | S_0 S_1 \rangle & \langle AC | H_{el} | CA \rangle & \langle AC | H_{el} | AC \rangle & & \\ \langle TT | H_{el} | S_1 S_0 \rangle & \langle TT | H_{el} | S_0 S_1 \rangle & \langle TT | H_{el} | CA \rangle & \langle TT | H_{el} | AC \rangle & \langle TT | H_{el} | TT \rangle \end{pmatrix} \quad (5)$$

where only the lower triangle part is shown because the matrix is Hermitian and the superscripts are omitted for simplicity. In this model, the singlet fission process (k_2 in Eq. (1)) is described as the

electronic transition from $|S_1S_0\rangle$ (or $|S_0S_1\rangle$) to $|TT\rangle$. We here assume that the molecules are equivalent and belonging to class I. Using the HOMOs (h_A, h_B) and LUMOs (l_A, l_B) for A and B, each electronic state (diabatic state, the basis of dimer) is described as follows [23].

$$\text{Singlet ground state} \quad S_0^A S_0^B = |h_A \bar{h}_A h_B \bar{h}_B| \quad (6a)$$

$$\text{Single excited state} \quad S_1^A S_0^B = \frac{1}{\sqrt{2}} \left[|h_A \bar{l}_A h_B \bar{h}_B| - |\bar{h}_A l_A h_B \bar{h}_B| \right], \quad (6b)$$

$$S_0^A S_1^B = \frac{1}{\sqrt{2}} \left[|h_A \bar{h}_A h_B \bar{l}_B| - |h_A \bar{h}_A \bar{h}_B l_B| \right] \quad (6c)$$

Correlated triplet pair state

$$\begin{aligned} {}^1(T_1^A T_1^B) = & \frac{1}{\sqrt{3}} \left[|h_A l_A \bar{h}_B \bar{l}_B| + |\bar{h}_A \bar{l}_A h_B l_B| \right. \\ & \left. - \frac{1}{2} \left\{ |h_A \bar{l}_A h_B \bar{l}_B| + |h_A \bar{l}_A \bar{h}_B l_B| + |\bar{h}_A l_A h_B \bar{l}_B| + |\bar{h}_A l_A \bar{h}_B l_B| \right\} \right] \quad (6d) \end{aligned}$$

$$\text{Singlet charge transfer state} \quad {}^1(C^A A^B) = \frac{1}{\sqrt{2}} \left[|h_A \bar{l}_B h_B \bar{h}_B| - |\bar{h}_A l_B h_B \bar{h}_B| \right], \quad (6e)$$

$${}^1(A^A C^B) = \frac{1}{\sqrt{2}} \left[|h_A \bar{h}_A l_A \bar{h}_B| - |h_A \bar{h}_A \bar{l}_A h_B| \right] \quad (6f)$$

Here, upper bar (non-bar) indicates β (α) spin orbital. These states are illustrated by Fig. 10. The excitation energies of these states are represented by $E(S_1S_0)$, $E(S_0S_1)$, $E({}^1T_1T_1)$ (${}^0E_{TT}$), $E({}^1CA)$, and $E({}^1AC)$, respectively. In the case of symmetric dimer composed of equivalent monomers, $E(S_1S_0) = E(S_0S_1)$ (${}^0E_{FE}$) and $E({}^1CA) = E({}^1AC)$ (${}^0E_{CT}$) are satisfied. Using Eqs. (6a–f), we can evaluate the matrix elements of H_{el} , which include Fock matrix elements and two-electron integrals using (h_A, h_B, l_A, l_B) [23,24]. Here, we obtain the analytical expression of transition probability from the initial state S_1S_0 to the final state ${}^1(T_1T_1)$ using perturbation theory in order to analyze the intermolecular interaction governing the transition probability.

Note that these electronic states Eqs. (10a–f) are not the eigenstates (adiabatic states) of the electronic Hamiltonian H_{el} that are evaluated by usual quantum chemistry calculations, but are diabatic basis. Diabatic basis have their own electronic characteristics such as local or charge transfer excitation. Diabatic and adiabatic basis can be transformed to each other, at least approximately, and thus are equivalent in the sense that both of them give the same spectrum of nuclear-electron Hamiltonian. Diabatic electronic states can be obtained by several theoretical techniques and have been applied to singlet fission studies, e.g., adiabatic-to-diabatic transformation [117–122], fragment molecular orbital calculation [24,30,97,102,123,124], orbital localization [109,114], constraining electron density [84], decomposition of an active space [96], non-orthogonal configuration interaction [125,126], projection onto spin Hamiltonian [127], and so on. We do not go in detail here. Note that, though electron correlation effects are apparently absent in the minimum configuration state functions model Eqs. (6a–f), it can be incorporated by evaluating effective diabatic states obtained from above theories. Adiabatic basis representation of singlet fission was also established [92,113,128–130], which has an advantage that the method in principle can treat any electronic states rather than shown in Eqs. (6a–f). We focus only on the diabatic basis approach here because of several reasons. First, diabatic basis are easy to understand because of their local and intrinsic properties. The intrinsic properties are substantially connected with the electronic structure of a monomer. Therefore, the diabatic basis approach is suited for materials design in the bottom-up approach. Second, diabatic representation is easy to use in dynamical simulation. As will be shown in Sec. 5, dynamical simulation including both electronic and nuclear degrees of freedom is very important to fully understand the mechanism of singlet fission in a given system. There fundamental formulation is usually done in diabatic basis. From these reasons, we consider singlet fission in diabatic representation. Of course, when the singlet fission process of a target system is highly complicated and is no longer described by a simplified model such as Eqs. (6a–f) or its extension, we should start from the adiabatic states description.

The first-order perturbation theory gives the singlet fission rate $w^{(1)}(\text{SF})$ (Fermi's Golden rule expression) expressed as [23,24]

$$w^{(1)}(\text{SF}) = \frac{2\pi}{\hbar} \left| \langle {}^1(\text{TT}) | H_{\text{el}} | S_1 S_0 \rangle \right|^2 \rho[E]. \quad (7a)$$

Here, $\rho[E]$ is the Franck-Condon-weighted density of states of ${}^1(\text{T}_1\text{T}_1)$ at $E = E_{\text{FE}} = E_{\text{TT}}$. This probability, which corresponds to the “direct” singlet fission pathway shown in Fig. 10, is found to be smaller in general than other off-diagonal matrix elements in the “indirect” pathways (see Fig. 10) [23,24,96,119] because the electronic coupling matrix is described by only two-electron integrals in the zero-differential overlap approximation as (for full representation including overlap matrix elements, see [124]),

$$\langle {}^1(\text{TT}) | H_{\text{el}} | S_0 S_1 \rangle = \sqrt{\frac{3}{2}} \left[\langle l_A l_B | h_B l_A \rangle - \langle h_A h_B | l_B h_A \rangle \right], \quad (7b)$$

where two-electron integrals indicate the electrostatic attracting/repulsion forces between the overlap densities between two chromophores. Since these overlaps are concerned with orbitals on different molecules, these overlaps are generally small. Also, in the zero-differential overlap approximation, this value becomes zero [24]. It was shown that in a slip-stacked dimer (which is slipped along the $h \rightarrow l$ transition), this value is maximized [23,24].

The next contribution comes from the second-order perturbation term, which indicates the indirect pathways shown in Fig. 10. Using quasi-degenerate perturbation theory of Hermitian form, the effective Hamiltonian matrix element $\langle \text{TT} | H_{\text{eff}} | S_1 S_0 \rangle$ for singlet fission considering up to the second-order contribution is expressed as [131,132]

$$H_{\text{eff}} = \langle {}^1(\text{TT}) | H_{\text{el}} | S_1 S_0 \rangle + \frac{1}{2} \sum_I {}^1\text{CA}, {}^1\text{AC} \left[\frac{\langle {}^1(\text{TT}) | H_{\text{ex}} | I \rangle \langle I | H_{\text{ex}} | S_1 S_0 \rangle}{E_{\text{FE}} - E_I} + \frac{\langle S_1 S_0 | H_{\text{ex}} | I \rangle \langle I | H_{\text{ex}} | {}^1(\text{TT}) \rangle}{E_{\text{TT}} - E_I} \right]. \quad (8a)$$

In the case of $E_{FE} = E_{TT}$, this expression is simplified and the singlet fission rate is given by

$$w^{(1)+(2)}(\text{SF}) = \frac{2\pi}{\hbar} \left| \left\langle {}^1(\text{TT}) \middle| H_{\text{el}} \middle| S_1 S_0 \right\rangle - \frac{\left\langle {}^1(\text{TT}) \middle| H_{\text{el}} \middle| {}^1\text{CA} \right\rangle \left\langle {}^1\text{CA} \middle| H_{\text{el}} \middle| S_1 S_0 \right\rangle + \left\langle {}^1(\text{TT}) \middle| H_{\text{el}} \middle| {}^1\text{AC} \right\rangle \left\langle {}^1\text{AC} \middle| H_{\text{el}} \middle| S_1 S_0 \right\rangle}{\Delta E_{\text{CT}}} \right|^2 \rho[E]. \quad (8b)$$

where $\Delta E_{\text{CT}} = E_{\text{CT}} - E_{FE} = E_{\text{CT}} - E_{TT}$. Each matrix element is represented by [23,24]

$$\left\langle {}^1(\text{TT}) \middle| H_{\text{el}} \middle| {}^1\text{CA} \right\rangle = \sqrt{\frac{3}{2}} \left[\langle l_A | F | h_B \rangle + \langle l_A l_B | h_B l_B \rangle - \langle l_A h_A | h_B h_A \rangle \right], \quad (9a)$$

$$\left\langle {}^1(\text{TT}) \middle| H_{\text{el}} \middle| {}^1\text{AC} \right\rangle = \sqrt{\frac{3}{2}} \left[\langle h_A | F | l_B \rangle + \langle h_A l_A | l_B l_A \rangle - \langle h_A h_B | l_B h_B \rangle \right], \quad (9b)$$

$$\left\langle {}^1\text{CA} \middle| H_{\text{el}} \middle| S_1 S_0 \right\rangle = \langle l_A | F | l_B \rangle + 2 \langle h_A l_A | l_B h_A \rangle - \langle h_A l_A | h_A l_B \rangle, \quad (9c)$$

$$\left\langle {}^1\text{AC} \middle| H_{\text{el}} \middle| S_1 S_0 \right\rangle = - \langle h_A | F | h_B \rangle + 2 \langle h_A l_A | l_A h_B \rangle - \langle h_A l_A | h_B l_A \rangle. \quad (9d)$$

In order to understand the dimer configuration dependences of the absolute value of the matrix elements in Eq. (8b), let us consider the stacked ethylene dimer model shown in Fig. 11. It is found that the matrix elements (Eqs. (9a–d)) are more than two orders of magnitude higher than those (Eq. (11b)) (see Figs. 18a and 19) because Eqs. (9a–d) include Fock matrix elements as well as a larger number of overlap densities on the same molecules in the two-electron integrals than those in Eq. (7b) [24]. Eqs. (9a–d) are involved in the second-order terms in Eq. (8b), but these terms are shown to disappear in the case of complete stack configuration due to the symmetry [23,24]. Therefore, similar to the first-order (direct path) contribution, the second-order (indirect path) contributions are found to be zero in the case of complete stack dimer. Fig. 13 shows orbital overlaps contributing to the Fock matrix elements, which are dominant in each matrix element. As seen from these diagrams, $\left\langle {}^1(\text{TT}) \middle| H_{\text{el}} \middle| {}^1\text{CA} \right\rangle$ (Eq. (13a)) increases positively from zero as increasing the slip distance along the

$h \rightarrow l$ transition dipole moment, takes a positive maximum when the slip distance is equal to the C–C bond distance, and then decreases toward zero value. In the present case, $\langle {}^1(\text{TT}) | H_{\text{el}} | {}^1\text{AC} \rangle$ (Eq. (9b)) has the same magnitude as $\langle {}^1(\text{TT}) | H_{\text{el}} | {}^1\text{CA} \rangle$ (Eq. (9a)) but has an opposite sign. On the other hand, $\langle {}^1\text{CA} | H_{\text{el}} | S_1S_0 \rangle$ (Eq. (9c)) takes the positive maximum in the complete stack case, decreases as increasing the slip distance, takes a negative extremum, and then increases again toward zero value. $\langle {}^1\text{AC} | H_{\text{el}} | S_1S_0 \rangle$ (Eq. (9d)) takes the negative extremum in the complete stack case, decreases toward zero value as increasing the slip distance. These features support the calculated results shown in Fig. 12. As seen from this figure, the second-order (indirect) pathway provides larger contribution to the singlet fission rate than the first-order (direct) one. Namely, singlet fission rate is predicted to be maximized in the slip-stack case with intermediate slip along the $h \rightarrow l$ transition dipole moment. In general, aromatic hydrocarbon compounds tend to form slip-stack structures in crystals, so that they are expected to be suitable for efficient singlet fission from the viewpoint of molecular configuration.

Finally, further approximation is applied to Eqs. (8) and (9) in order to more clarify the molecular configuration effects on the amplitude of singlet fission transition matrix. Under the approximation of zero-differential overlap between orbitals on different molecules, the first-order (direct) term disappears and only the Fock matrix elements remain in the second-order (indirect) terms. Eq. (8b) is thus approximately expressed as

$$w^{\text{ap}}(\text{SF}) = \frac{3\pi}{\hbar} \left| \frac{\langle l_A | F | h_B \rangle \langle l_A | F | l_B \rangle - \langle h_A | F | l_B \rangle \langle h_A | F | h_B \rangle}{\Delta E_{\text{CT}}} \right|^2 \rho[E]. \quad (10)$$

Here, if the dimer has a symmetry such as inversion symmetry and plane symmetry,

$\langle l_A | F | h_B \rangle = - \langle h_A | F | l_B \rangle$ is satisfied as seen from Fig. 13 (be care for the relative sign of MOs), so

that Eq. (10) is further reduced to

$$w^{\text{ap}}(\text{SF}) = \frac{3\pi}{\hbar} \left| \frac{\langle l_A | F | h_B \rangle (\langle l_A | F | l_B \rangle + \langle h_A | F | h_B \rangle)}{\Delta E_{\text{CT}}} \right|^2 \rho[E]. \quad (11)$$

There are many experimental and theoretical studies that show the importance of electronic coupling in singlet fission rate and yield, e.g. different crystal forms of diphenylhexatriene derivatives [48,87], 1,3-diphenylisobenzofuran [88,89], tetracene [90–94,98] and pentacene aggregates [84,92,95,97,119], perylene-3,4-dicarboximide dimer model [99], slip-stacked perylenediimide derivatives [100–102], covalently-linked tetracene and pentacene dimers [30,104–111,16,112–116]. The singlet fission rate constant can differ by several orders of magnitude even in slightly different but with essentially the same core chromophore [84]. These results provide clear evidence that molecular configuration affects the rates of singlet fission. As the electronic coupling is originated from orbital overlap at a pair of neighboring chromophores, local environment [94,133,134] can be significantly important for singlet fission in non-crystalline solids.

4.2. Singlet fission in pancake-bonded systems

We here introduce a novel class of singlet fission systems called “pancake-bonded” systems. So far, most of the observed and proposed singlet fission aggregates are limited to usual closed-shell, or weak diradical, molecular aggregates. On the other hand, there have been a unique class of molecular aggregates including π -stacked (mono-)radical dimer, which create a covalent-like bonding between the monomers, referred to as “multi-center two-electron bonding” or “pancake bonding” [135–141]. As an example, we consider phenalenyl radical (**Phn**, Fig. 14a), which is known to create a covalent-like bonding in its dimer **Phn₂** and aggregates through the bonding interaction between the singly occupied molecular orbitals (SOMOs) of each monomer [136,139–144]. The SOMO of **Phn** is shown to be delocalized in the whole region of the molecule on α carbon atoms (see Fig. 14a and b). Two forms of π -dimers, staggered and eclipse forms, are known to be synthesized using different substituents [136,139,142,144–147]. Large π -orbital overlap in pancake-bonded systems promotes

intensive investigation of many physical properties [140,148–150]. In addition to these unique and wide-applicable functionalities of open-shell aggregates, we here introduce another potential application to singlet fission [151].

As discussed in the previous Section 4.1, the slipped-stack configuration along the direction of transition dipole moment of a monomer is predicted to be promising for efficient singlet fission, while this slip is also predicted to reduce the horizontal couplings F_{HH} and F_{LL} that are required to be large for charge transport. Thus, due to the trade-off relationship, the SF and charge-transport rates are not expected to be maximized simultaneously. For the application of singlet fission to a photovoltaic cell, charge-transport property should not be abandoned. However, this trade-off relationship between SF and charge-transport is not found to hold for pancake-bonded systems as shown later. Therefore, using pancake-bonded systems for singlet fission application is a highly appealing design guideline.

Electronic couplings of a potential pancake-bonding system, phenalenyl tetramer **Phn₄**, and of conventional chromophore, pentacene dimer **Pc₂**, are shown in Fig. 15. Note that here a phenalenyl radical is considered to constitute a singlet chromophore as a dimer. Although **Pc₂** shows zero effective coupling V_{SF} at the slipped-stack configuration, Phn4 does significant amplitude of V_{SF} at the same configuration. This was explained by the characteristic nodal pattern in molecular orbitals found in pancake-bonding systems. Having the largest electronic coupling V_{SF} , F_{HH} and F_{LL} at the face-to-face configuration, pancake-bonding systems are potential class of singlet fission materials with high charge mobility.

From Eqs. (5) and (9), the effective Hamiltonian in the quasi-degenerate second-order perturbation theory for a symmetric singlet fission dimer is expressed by [131,151]

$$\begin{pmatrix} S_1 S'_0 \\ S_0 S'_1 \\ TT' \end{pmatrix} \begin{pmatrix} E_{FE}^{eff} \\ J^{eff} & E_{FE}^{eff} \\ V_{SF}^{eff} & V_{SF}' & E_{TT}^{eff} \end{pmatrix} = \begin{pmatrix} E_{FE} + DE_{FE} \\ J_{Coul} + J_{CT} & E_{FE} + DE_{FE} \\ V_{SF} & V_{SF}' & E_{TT} + DE_{TT} \end{pmatrix}, \quad (12)$$

where

$$DE_{FE} = \frac{F_{HH}^2}{E_{FE} - E_{CT}} + \frac{F_{LL}^2}{E_{FE} - E_{CT}} \quad (13a)$$

$$DE_{TT} = \frac{(\sqrt{3/2}F_{LH})^2}{E_{TT} - E_{CT}} + \frac{(\sqrt{3/2}F_{HL})^2}{E_{TT} - E_{CT}} \quad (13b)$$

$$J_{CT} = 2 \frac{-F_{HH}F_{LL}}{E_{FE} - E_{CT}} \quad (13c)$$

$$V_{SF} = \frac{1}{2} \left(\sqrt{\frac{3}{2}} \frac{-F_{HH}F_{HL} + F_{LL}F_{LH}}{E_{FE} - E_{CT}} + \sqrt{\frac{3}{2}} \frac{-F_{HH}F_{HL} + F_{LL}F_{LH}}{E_{TT} - E_{CT}} \right) \quad (13d)$$

$$V_{SF}' = \frac{1}{2} \left(\sqrt{\frac{3}{2}} \frac{-F_{HH}F_{LH} + F_{LL}F_{HL}}{E_{FE} - E_{CT}} + \sqrt{\frac{3}{2}} \frac{-F_{HH}F_{LH} + F_{LL}F_{HL}}{E_{TT} - E_{CT}} \right). \quad (13e)$$

The primes on the basis state symbols indicate that they are the effective basis including the CT-mixing effect. The Coulomb and CT-mediated contribution in the excitonic coupling, J_{Coul} and J_{CT} , are known to make a strong impact on optical property of organic semiconductors [152,153].

The effective energies and those split by the effective exciton-exciton coupling J^{eff} of **Phn₄** and **Pc₂** are shown in Fig. 16. **Phn₄** exhibits modest energy change by the CT-mixing energy perturbations ΔE_{FE} and ΔE_{TT} for the displacement along the x-axis. In **Phn₄**, the exchange of the relative energy order of FE' and TT' is not found even after the splitting by J^{eff} . Namely, all the Fock matrix elements vary in the same manner in the whole configuration (x) region. As long as the relative amplitude of the Fock matrix elements is the same at each geometry, the perturbation effects on the energy and couplings are found to vary in the same manner, and hence do not change

the relative energy balance drastically. The lowest state in **Phn₄** is shown to keep significant TT character in the whole x region, resulting in high-yield SF in a wide range of configurations. In contrast to **Phn₄**, in **Pc₂**, due to the individual variation of the Fock matrix elements as well as to the significant oscillation of the CT-mixing energy perturbations, $E_{\text{FE}}^{\text{eff}}$ of **Pc₂** is found to be higher than $E_{\text{TT}}^{\text{eff}}$ only in $x > 3.0$ Å. These findings stimulate application of pancake bonding to singlet fission owing to its high singlet fission potential from the viewpoint of energetics and electronic couplings, as well as to its high charge-carrier transport property.

4.3. Intramolecular singlet fission in covalently-linked systems: singly-bridged systems

In addition to the crystal engineering on relative configurations of chromophores for efficient SF, there is another possible way to modify the electronic coupling, where two chromophores are covalently linked via a bridge [104–111,116,112,113]. Early attempts for singlet fission in covalently-linked systems composed of tetracenes definitely indicated a generation of triplet-pair exciton in a dimer, while its rate was low (hundreds of nanoseconds) and triplet yield was poor [104,105]. A very high SF efficiency up to 156% in 10^1 – 10^3 ps time scales, however, has been experimentally observed in covalently-linked systems composed of two pentacenes linked via *ortho*-, *meta*- and *para*-diethynylbenzenes (referred to as **o-Pc**, **m-Pc** and **p-Pc**, respectively, Fig. 17a–c), respectively [111]. Because pentacene is found to occur exothermic SF due to $2E(T_1) - E(S_1) = -0.11$ eV [154,155], the difference in the electronic coupling is predicted to cause the different SF dynamics in these systems. In this section, we theoretically investigate covalently-linked systems with different linkers shown in Fig. 17a–c to clarify the difference in their electronic couplings [115]. The results contribute to constructing design principles for realizing moderate electronic couplings suitable for efficient SF in covalently-linked systems. Then, on the basis of the design principles, we propose several new covalent bridges and verify their performance.

The SF electronic coupling is expressed as Eq. (9). The Fock matrix elements F_{ij} are evaluated by using a non-orthogonal effective Hamiltonian, or Green's function, approach [156], where the Fock matrix elements are described as matrix elements of the effective Fock operator:

$$\hat{P}^\dagger \hat{F}^{\text{eff}} \hat{P} = \hat{P}^\dagger (\hat{F} - E\hat{I}) \hat{P} + \hat{P}^\dagger (\hat{F} - E\hat{I}) \hat{G}_{\hat{Q}^\dagger \hat{Q}} (\hat{F} - E\hat{I}) \hat{P}. \quad (14)$$

Here, $\hat{F}$ and $\hat{F}^{\text{eff}}$ are the original and effective Fock operators; $\hat{I}$ is the identity operator; P and Q are projection operators that project the one-electron orbital space onto the HOMO and LUMO of the chromophores, and onto the other MOs, respectively; $\hat{G}_{\hat{Q}^\dagger \hat{Q}}$ is the Green's function depending on the Fermi energy E . The Green's function approach allows us to partition each matrix element F_{ij} into two types of contributions, "direct-overlap" and "bridge-mediated" couplings, which are approximately the first and the second terms in Eq. (14), respectively. The former and latter represent the direct and indirect interactions between π -orbitals of the chromophores, respectively.

Here, we discuss the differences in SF electronic couplings between **o-**, **m-** and **p-Pc**. As seen from the electronic couplings (Table 2), the direct-overlap couplings are the largest in **o-Pc** and they are two or three orders of magnitude larger than those of **m-Pc** and **p-Pc**. This feature is predicted to be caused by the significant direct π -orbital overlap between the chromophores of **o-Pc** (see Fig. 17a). It is also found that the bridge-mediated couplings have two or three order larger amplitudes in **p-Pc** and **o-Pc** than in **m-Pc**. Thus, the amplitudes of total electronic couplings F_{ij} are found to decrease in the order: **o-Pc** > **p-Pc** > **m-Pc**. The same ordering is also observed in the SF coupling amplitude $|V_{\text{SF}}|$. Note here that this tendency of $|V_{\text{SF}}|$ is in qualitative agreement with the SF rate observed by a transient spectroscopy [111].

In order to clarify the origin of the difference in the electronic couplings in **o-**, **m-** and **p-Pc**, we apply physically appropriate approximations for Eq. (14). The formula for the bridge-mediated term is expressed by [115]

$$F_{ij}^{\text{Bridge-mediated}} \approx c_m c_{n'j} b^2 \left(\frac{c_{m_{HB}} c_{n_{HB}}}{E - E_{HB}} - \frac{c_{m_{LB}} c_{n_{LB}}}{E_{LB} - E} \right). \quad (15)$$

Here, b is the resonance integral; c is a MO coefficient; subscripts i and j indicate the H1, L1, H2, or L2; HB and LB represent the HOMO and LUMO of the bridge with orbital energies E_{HB} and E_{LB} , respectively; subscripts 1, 2 and B represent the chromophore 1, 2 and the bridge, respectively; Greek subscripts with and without prime indicate a pair of carbon atoms, where a fragment pair (1 and B) or (2 and B) is linked with the chemical bond μ' - μ or ν' - ν ; E denotes the Fermi energy level. In Eq. (15), the denominators are positive because the Fermi energy E is shown to lie in between E_{HB} and E_{LB} . The two terms in the parenthesis indicate that the relative phase between the HB and LB is crucial for moderate bridge-mediated coupling, i.e., the different (the same) relative sign of the products in the numerators leads to constructive (destructive) contribution. In alternant hydrocarbons bridges, the relative phase of the HB and LB at the linked carbon atoms is determined by Coulson-Rushbrooke pairing theorem [157], which states that only the starred-carbon atoms change their MO coefficients in sign between the HOMO and LUMO. On the basis of this theorem, the bridge-mediated coupling becomes large when two chromophores are linked at starred and non-starred carbons of a bridge, while that does small when linked at two starred (or two non-starred) carbons of a bridge. Because **o-Pc** and **p-Pc** are classified into the former, they are predicted to have opposite sign at the linked carbon in the HB and LB. In contrast, because **m-Pc** is classified into the latter, it is predicted to have the same sign. We can confirm the above discussion by decomposing the electronic couplings into each MO contribution. In Fig. 18, we find that the above simple considerations work very well even with *ab initio* methods.

On the basis of these results, new design principles for efficient intramolecular SF are presented in addition to the previous two types of bridges, i.e., bridges putting two chromophores in a small distance such as **o-Pc**, and using alternant hydrocarbon bridges linked at starred and non-starred carbons such as **p-** and **o-Pc**. Furthermore, we present three additional design strategies for

potential bridges: non-alternant hydrocarbons, bridges including heteroatoms and radicals. These are all derived from the consideration of the interference effect principle described by Eq. (15). Note here that these situations are not exclusive with each other. Using these principles, we here design three types of bridges, **Az-Pc**, **Nm-Pc** and **Rad-Pc**, shown in Fig. 17d–f. These bridges are found to cause a wide range of V_{SF} (1–138 meV) (see Table 2). The mechanisms causing moderate electronic coupling are understood by checking the shape and energy of MOs in the bridges [115]. In the bridge of **Az-Pc**, the LB has significant amplitude at the linked carbon atoms, while the HB has zero amplitude due to a node at the carbon atoms. This leads to significant and zero contributions in the bridge-mediated coupling from LB and HB, respectively, i.e., moderate V_{SF} . In **Nm-Pc**, the breaking of orbital energy symmetry is found to be the key for enhancement in two orders in magnitude of V_{SF} , from 0.01 meV to 1 meV. In **Rad-Pc**, small energy difference in the denominator is found to cause the moderate V_{SF} .

4.4. Intramolecular singlet fission in covalently-linked systems: doubly-bridged systems

Covalently-linked systems can be extended to multiply-bridged systems where two or more bridge moieties link two chromophores. This, as shown below, paves a way of rational control of dimeric electronic state energies through electronic coupling design. The simplest case of multiply-bridged systems is a doubly-bridged chromophore dimer, and we focus on it. The bridge-mediated coupling in a doubly-bridged system is approximately described in a similar form with Eq. (15) with the sum for bridges,

$$F_{ij}^{\text{eff bridge-mediated}} \approx \sum_{b \text{ all bridges}} c_{\mu(b)'} c_{\nu(b)'} j \beta^2 \left(\frac{c_{\mu(b)\text{HB}} c_{\nu(b)\text{HB}}}{E - E_{\text{HB}(b)}} - \frac{c_{\mu(b)\text{LB}} c_{\nu(b)\text{LB}}}{E_{\text{LB}(b)} - E} \right), \quad (16)$$

where b is the index for bridges ($b = 1, 2$). As seen from Eq. (16), both the relative phases of prefactor in the front of the parentheses and of the numerators in the parentheses are significantly important for the bridge-mediated coupling because of the interference effects from different

bridges, $b = 1$ and 2, and within the bridges. Especially, the prefactors, which are related to the phase of chromophore MOs at the bridged atoms and are not focused in the previous section, play a crucial role in doubly- (multiply-)bridged systems.

From the consideration of phases in MOs, we can classify the interference patterns into four, in symmetric dimer systems with two constructive bridges. These four cases are derived from two cases of bridging positions, (I) head-to-head and (II) head-to-tail transition dipole moment configurations, and from two cases of intrinsic nature in the MO phase of a bridge, positively-constructive (PC) and negatively-constructive (NC) choices of bridges. The nature of a constructive interfering bridge, which is classified into PC and NC, is determined from the relative phase of the HOMO and LUMO of the bridge. For example, when the HOMO of a bridge corresponds to a symmetric superposition of atomic orbitals while the LUMO of a bridge does to an antisymmetric one, the parentheses term in Eq. (16) for this bridge is positive, which represent a PC bridge, and *vice versa*.

We now go to the relationship between the above consideration and singlet fission. The above essential four cases, that is, (I-PP), (I-PN), (II-PP) and (II-PN), are schematically shown in Fig. 19. PC and NC bridges are denoted by solid and broken lines, respectively. In the case (I-PP), in all the Fock matrix elements F_{ij} , the quantum interference between bridges is constructive, leading to large electronic coupling amplitudes in all the elements. In the case (I-PN), the situation is opposite: in all the Fock matrix elements F_{ij} , the quantum interference between bridges is destructive, leading to small electronic coupling amplitudes in all the elements. In the case (II-PP), due to the PC nature of the bridges 1 and 2 and mutually anti-phase HOMO-LUMO relation of chromophores at the two linked sites, constructive interference in “horizontal” couplings (F_{HH} and F_{LL}) results in large amplitudes of them, while destructive interference in “non-horizontal” couplings (F_{HL} and F_{LH}) results in small amplitudes of them. In the case (II-PN), the situation is opposite to the previous case: destructive interference in horizontal couplings results in small amplitudes of them, while

constructive interference in non-horizontal couplings results in large amplitudes of them. In short, it is found that in the case (I-PP) (in the case (I-PN)), all the Fock matrix elements are large (small) in their amplitudes; in the case (II-PP) (in the case (II-PN)), the horizontal couplings are large (small) in their amplitudes, while the non-horizontal couplings are small (large) in their amplitudes.

Owing to different amplitudes in the horizontal and non-horizontal couplings in the above four cases, the doubly-bridged systems should have also significantly different effective excitation energies even when the same chromophores are considered, Eqs. (12). Recall that large horizontal couplings stabilize FE states, while large non-horizontal couplings stabilize TT state (Eqs. (13)). This consideration strongly suggests that the electronic state energies can be controlled through the configuration interaction by controlling electronic coupling in desired ways. This is crucial for singlet fission because the relative energy balance between FE and TT states are recognized as the most important issue.

The above considerations are applied for applied for tetracene dimers bridged with two π -conjugated bridges, n -polyyne ($-(C\equiv C)_n-$) (Fig. 20). As shown in the example of ethylene and butadiene bridges, odd and even numbered polyyne are expected to work as positively and negatively constructive bridges, respectively. From the consideration in the relative phase of the HOMO and LUMO, tetracene is expected to work as the cases (I-PP) or (I-PN) when two bridges are linked at the 1- and 10-positions (**Tc1-(m,n)**), while as the cases (II-PP) or (II-PC) when linked at 2- and 3-positions (**Tc2-(m,n)**). The evaluated electronic couplings are summarized in Table 1. As expected, the four cases, (I-PP), (I-PN), (II-PP) and (II-PN), are realized in the even combinations ($m + n = \text{even}$) of **Tc1-(m,n)**, odd combinations of **Tc1-(m,n)**, even combinations of **Tc2-(m,n)**, and odd combinations of **Tc2-(m,n)**, respectively. The predicted interference effect between bridges are definitely confirmed by decomposition of electronic couplings into direct-overlap and bridge-mediated contributions from the bridge 1 and 2, see Fig. 21. There contributions in electronic couplings from bridges change in their signs for odd and even parity of bridge length n . This confirms that the even

combinations are the cases (PP), while the odd combinations are the cases (PN). The signs of the contributions from the bridges 1 and 2 in horizontal and non-horizontal couplings are the same in **Tc1-(*m,n*)**, while opposite in **Tc2-(*m,n*)**. This clearly indicates the former cases are the cases (I) and the latter the cases (II). The effective energies are found to be significantly different for each case, see Table 1. Owing to the large difference between the horizontal and non-horizontal couplings, the case (II-PP) shows much lower energy in the FE' state than that in the TT' state, while the case (II-PN) shows much lower energy in the TT' state than that in the FE' state. The results strongly suggest rationally controlled electronic couplings pushes the effective TT state energy down significantly and can make singlet fission energetically feasible. Focusing on the other way of utilizing our strategy, the reverse process of singlet fission, triplet–triplet annihilation, can be enhanced through the stabilization of the FE state.

We have shown the importance of a rational design of electronic couplings and of the potential for the effective energy tuning. Such rational design is definitely performed and is shown to be possible by using carefully choosing linking positions and bridge moieties. Further investigations of covalently-linked systems both from experimental and theoretical studies are needed for deeper understanding in the mechanisms and for improving the applicability of them.

5. Exciton dynamics in singlet fission

5.1. Vibronic coupling

The interaction between electronic and nuclear vibrational motion is referred to as vibronic coupling. In the electronic ground state, nuclei in a molecule have zero-point vibrations around the equilibrium positions. When the optical excitation occurs from the electronic ground state to an electronic excited state, of which potential minimum is different from that of the electronic ground state, the nuclear motions are induced along the potential minimum of the excited state. This is an

example of vibronic coupling or electron–phonon coupling. As will be described in detail below, there are two types of vibronic couplings, i.e., Holstein coupling and Peierls coupling [158]. The Holstein coupling causes a fluctuation of electronic energy (diagonal term in the electronic coupling matrix, so that it is often referred to as “local electron-phonon coupling”), while the Peierls coupling does a fluctuation of electronic coupling (off-diagonal term in the electronic coupling matrix, so that it is often referred to as “nonlocal electron-phonon coupling”). Namely, the Holstein coupling corresponds to the structure relaxation caused by the electronic state change, and often relates to the bond stretching mode. In singlet fission, the Holstein coupling causes the fluctuation of the energy gap between the optically-allowed state S_1 and forbidden state $^1(TT)$, which is thus predicted to affect the energy matching conditions dynamically. On the other hand, the Peierls coupling, which is mostly a function of intermolecular modes, is related to the non-adiabatic transition including non-radiative decay.

Such nuclear motions are theoretically described by two approaches, i.e., reaction coordinate search [39,80,91,92,119,120,125,159,160] and linear vibronic coupling model approach [29,30,97,102,114,123,161–172]. In the former approach, we choose a few (one or two) reaction coordinates, and investigate the energy variation along these coordinates. This approach has advantages, e.g., a reduction of the vibration motion to a few number of degree of freedom gives easily comprehensible interpretations of the mechanism, and a possible full quantum description of vibrational motions including anharmonicity, while this has deficiencies, e.g., difficulty in getting prior knowledge about reaction coordinates, high computational costs to obtain precise potential surfaces, and a possibility of insufficient number of degree of freedom for the description of real vibrational

motions. Zimmerman et al. theoretically investigated singlet fission in tetracene and pentacene crystals by using this approach and found two important features for singlet fission: (1) localization of S_n (the n th singlet excited state) states via formation of an excimer-like state, and (2) coupling along the same intermolecular coordinate which leads to rapid population of the D state (double excited state) [159,39]. Fig. 22 shows the proposed mechanism of singlet fission (a) and calculated potential curves (b) together with the model dimers. They found that in acene crystals, localization of the S_1 exciton through intermolecular motion facilitates state crossing to a localized TT state on a dimer. The increase of the π overlap between neighboring monomers in pentacene leads to a near-degeneracy between S_1 and D, resulting in efficient singlet fission. In contrast, in tetracene, the downhill non-adiabatic transition from S_3 into D does not appear to occur through excited-state dimers, while vibrational excitation along the excited-state dimer coordinate can explain the uphill pathway from S_1/S_2 to D. Zeng et al. performed high-level electron correlation calculation for a pentacene dimer and investigated its potential energy curve along intermolecular motions [119]. They suggested that intermolecular motion can change diabatic state energies drastically, especially for a charge transfer state. On the other hand, in a covalently-linked pentacene, intramolecular vibrations are found to be important for the energy crossing between the FE and TT states [120]. Physically meaningful reaction coordinate(s) can be qualitatively different by chromophores [80]. These imply that we may not be able to capture the essential energy landscape of singlet fission in a single reaction coordinate model. We will have to abandon calculation of multidimensional potential energy surface when the required dimension goes up to three or more due to high computational

cost and difficulty in its interpretation. We may adopt the linear vibronic coupling approach for such a situation.

The linear vibronic coupling approach treats the information of multi-dimensional potentials by coupling constants between the electronic states and vibrational degrees of freedom. This approach has advantages, e.g., possible microscopic analysis based on the contribution of each vibration mode, and an easy treatment of wavepacket or density matrix dynamics by using a linear vibronic coupling Hamiltonian, while this has difficulty in the description of anharmonic vibrational effects. Still, higher-order contribution in vibronic couplings can be evaluated and the effect can be examined in dynamical simulations. Such higher order effect may be important for highly flexible molecules, though it is not often seen in the main π -backbones in singlet fission chromophores. In the (linear or higher) vibronic coupling Hamiltonian, the electronic Hamiltonian is expanded as a Taylor series around the equilibrium point $\mathbf{Q} = \mathbf{Q}_0$ of the reference state (usually the ground state),

$$H_{mn}(\mathbf{Q}) = H_{mn}(\mathbf{Q}_0) + \sum_i \hat{Q}_i \left. \frac{\partial H_{mn}(\mathbf{Q})}{\partial Q_i} \right|_{\mathbf{Q}=\mathbf{Q}_0} + \sum_{i,j} \hat{Q}_i \hat{Q}_j \left. \frac{\partial^2 H_{mn}(\mathbf{Q})}{\partial Q_i \partial Q_j} \right|_{\mathbf{Q}=\mathbf{Q}_0} + \dots, \quad (17)$$

where m and n are electronic states, i and j are vibration modes of the ground state, and $\{Q_i\}$ denote dimensionless vibrational coordinates. The first term (zeroth-order term) in Eq. (17) represents electronic energy ($H_{mn}(\mathbf{Q}_0)$), where $n = 1$ and $n > 1$ indicate the ground state and vertical excitation energies in the diabatic basis, respectively) and coupling ($H_{mn}(\mathbf{Q}_0)$ ($m \neq n$)) at $\mathbf{Q} = \mathbf{Q}_0$. The second term (the first-order term) indicate the linear vibronic coupling, which has two types of couplings: Holstein coupling (HC, $m = n$) and Peierls coupling (PC, $m \neq n$). The former indicates the fluctuation of energy of the n th state ($n > 1$), while the latter the fluctuation of electronic coupling between states m and n . The third term (the second-order term) with $m = n$ includes the energy of phonon

bath for electronic state n , and the interaction of vibration modes i and j for electronic state n . This Hamiltonian has been widely applied to many vibronic problems in ultrafast phenomena and transport phenomena [173–179]. In the linear vibronic coupling Hamiltonian for singlet fission study, we consider Holstein and Peierls couplings, and phonon bath energy. In most singlet fission studies, Peierls coupling has been ignored due to the much smaller amplitudes than the Holstein coupling, and a common phonon bath for each electronic state is considered. However, recent theoretical studies have pointed out that neglecting Peierls couplings and the state dependence of phonon bath are inappropriate for the description of singlet fission [27,123,160,167,171,180]

5.2. Vibronic coupling density analysis

In order to analyze and evaluate the relationships between vibronic coupling and molecular geometric/electronic structure, we here consider the vibronic coupling density (VCD) [181–184]. The first-order VCs are shown to be related to spatial functions as

$$\left. \frac{\partial H_{mn}(\mathbf{Q})}{\partial Q_i} \right|_{\mathbf{Q}=\mathbf{Q}_0} = \nabla \mathbf{r} r_n^{\text{diff}}(\mathbf{r}) v_i(\mathbf{r}) \quad (18a)$$

$$\left. \frac{\partial H_{mn}(\mathbf{Q})}{\partial Q_i} \right|_{\mathbf{Q}=\mathbf{Q}_0} = \nabla \mathbf{r} r_{mn}^{\text{tra}}(\mathbf{r}) v_i(\mathbf{r}). \quad (18b)$$

Here, $r_n^{\text{diff}}(\mathbf{r})$ indicates difference density defined by

$$r_n^{\text{diff}}(\mathbf{r}) = r_n(\mathbf{r}) - r_0(\mathbf{r}), \quad (19)$$

where $r_n(\mathbf{r})$ denotes the electron density of electronic state n ($n = 0$ is the ground state). A

potential derivative $v_i(\mathbf{r})$ is defined as

$$v_i(\mathbf{r}) = \sum_A^{N_{\text{atom}}} \left(\frac{\partial}{\partial Q_i} \left(\frac{-Z_A}{|\mathbf{r} - \mathbf{R}_A|} \right) \right) \bigg|_{\mathbf{Q}=\mathbf{Q}_0} \quad (20)$$

The integrands on the right-hand sides of Eqs. (18a) and (18b) are referred to as VCDs for HC and PC, respectively, since their spatial integrals give the corresponding VCs. Because VCDs can be separated into two parts, $r_n(\mathbf{r})$ ($r_{mn}^{\text{tra}}(\mathbf{r})$) and $v_i(\mathbf{r})$, we can reveal the origin of VCs from the viewpoint of electronic and vibrational structures by the VCD analysis. In order to clarify the performance of VCD analysis, let us consider a Holstein VC induced by a C-C stretching mode of naphthalene upon the HOMO $\rightarrow$ LUMO singly excitation, see Fig. 23. The potential derivative distribution $v(\mathbf{r})$ for this mode and the electron density difference (the HOMO-LUMO density difference in this case) distribution $D(\mathbf{r})$ are key variables for understanding the VC of this vibration mode. The VCD for HC, which is defined by the product of $v(\mathbf{r})$ and $D(\mathbf{r})$, is illustrated in Fig. 23c. The VC, the integral of the VCD, is shown to be negative in sign in this case. This implies that the HOMO $\rightarrow$ LUMO excitation causes the nuclear motion along this C-C stretching mode toward its excited state equilibrium structure.

Let us go to VC in singlet fission. We consider a slip-stack tetracene dimer model composed of monomers A and B, which has an intermolecular distance of 4.0 Å with 0.5 Å shift in the lateral direction as shown in Fig. 24a [162]. We here consider five diabatic excited electronic states of Eq. (5), which are schematically shown in Fig. 24b. In VC calculation in the dimer system, there are two ways of the superposition of each monomer vibration mode, that is, “in-phase” and “out-of-phase” superposition in the dimer. As examples, we here explain only the HCs (see [162] for PCs).

Fig. 25 shows the calculated the amplitudes of VCs (see Eq. (18)) of FE1 (FE2) (a), CT1 (CT2) (b), and TT (c) states. We discuss selected vibration modes indicated by arrows as examples. Hereafter, “in-phase” and “out-of-phase” modes are expressed by adding “i” and “o” superscripts on the frequency, respectively. It is found that there are significant differences in overall amplitudes and characteristic frequencies giving large HCs, both of which depend on the electronic states and the relative phase (“in-phase” or “out-of-phase”) of vibrational motion in each monomer. The relative phase dependence is found for CT1 and TT states, while not for FE1 state. FE1 state shows

the three largest HC peaks (263 meV, 268 meV, and 267 meV), which correspond to a ring-breathing mode ($1271^{\text{i},\text{o}} \text{ cm}^{-1}$), and two carbon-carbon (C-C) stretching modes ($1455^{\text{i},\text{o}} \text{ cm}^{-1}$ and $1651^{\text{i},\text{o}} \text{ cm}^{-1}$), respectively. CT1 state shows the three largest HC peaks (721 meV, 1411 meV, and 1005 meV), which correspond to an out-of-plane bending mode ($60.5^{\text{i}} \text{ cm}^{-1}$) and two ring-breathing modes ($328^{\text{o}} \text{ cm}^{-1}$ and $794^{\text{o}} \text{ cm}^{-1}$), respectively. TT state shows the three largest HC peaks (though only in in-phase modes) at the same frequencies as those in FE1 state. Interestingly, the HC intensities of TT state are shown to be twice the FE1 ones, that is, 526 meV (TT) vs. 263 meV (FE1) at $1271^{\text{i}} \text{ cm}^{-1}$, 535 meV (TT) vs. 268 meV (FE1) at $1455^{\text{i}} \text{ cm}^{-1}$, and 533 meV (TT) vs. 267 meV (FE1) at $1651^{\text{i}} \text{ cm}^{-1}$.

Using the vibronic coupling density (VCD) analysis, now we reveal the microscopic origin of these VCs. Fig. 26 shows the difference densities of FE1 (a), CT1 (b) and TT (c) concerning HCs. The VCDs are obtained by multiplying these densities by the potential derivatives with respect to the corresponding normal mode coordinate, and the corresponding VCs are calculated by the spatial integral of the VCDs. The difference density is proportional to the difference between the unoccupied and occupied MO densities concerning the excitation. The difference densities of FE1 and TT states show negative distributions for the C-C double bonds due to their π - π^* excitation nature, i.e., bonding to anti-bonding excitation, typically observed in alternant hydrocarbons, though in TT state, the same difference density distribution is observed in both molecules due to its two-electron excitation nature. For CT1 state, the difference density shows positive and negative distributions on the upper and the lower molecules, respectively, due to its charge transfer nature. The VCDs are characterized by the overlap distributions between these densities and potential derivatives (Fig. 27). Because the potential derivative of the C-C stretching mode at $1455^{\text{i}} \text{ cm}^{-1}$ (Fig. 27d) shows alternating positive and negative distribution between each C-C bond, which well overlaps with the difference density of TT state with the same sign in a common distribution region, large HC is obtained in TT state. Similar difference density distribution in monomer is observed in FE1 and TT states (Fig. 26), excepting that FE1 and TT have the distribution in only the upper monomer and in both monomers, respectively. Thus, for in-phase vibrations, the HC peak intensities

of TT state (Fig. 25c) are found to be almost twice those of FE1 state (Fig. 25a). This doubled-value of HCs in TT as compared to those of FE1 indicates that such high-frequency C-C stretching modes can be an effective reaction coordinate because these modes should reduce the energy difference between FE1 and TT states and make them near-degenerate. Indeed, the present result is in agreement with the recent observations of vibronic coherence in ultrafast timescale [27,32], and more sophisticated theoretical calculations [91,120,172].

VCs could be designed in rational ways from the insights through Eq. (29) [183,184]. For example, obtaining large HCs may be achieved by putting donor and acceptor moieties in a chromophore, because strong charge transfer nature in the local excitation (FE) should induce large difference density (Eq. (18a)). Reduction of the molecular symmetry also induces new source for HCs that is absent in the high-symmetry parent chromophore. Alternate hydrocarbons such as tetracene are not the cases shown above, thus HCs are not so large. Chemical modifications above should not break the energy level matching conditions for application of singlet fission, of course. For off-diagonal parts of VCs, i.e., PCs, large orbital overlap between chromophores could be a key for enhancing them. This is because PCs are related to transition densities (Eq. (18b)), which are large in their amplitudes at the regions with large orbital overlap. Note that an electronic coupling matrix is not needed to be large, because PC are dominated by the differential overlap, not the integrated one. In this direction, we soon come up with a complete stack dimer as a potential large PC system, though its electronic couplings related to TT themselves are zero. This may be the case for rubrene. Although rubrene shows high charge mobility [185], its electronic couplings related to TT are zero by symmetry [180]. Both theoretical and experimental studies support the significant effect of PC for this system [160,180]. Even when PCs are large, however, the effect of PCs may be concealed by large electronic couplings, which is also large when the orbital overlap is large. Recent experimental and theoretical studies enlightened significant effect of both HCs and PCs on ultrafast dynamics of singlet fission [27,32,91,119,120,123,159,160,166–168,180]. Those studies highly motivate one to explore rational routes for vibronic coupling design toward efficient singlet fission materials.

5.3. Theory of exciton dynamics, including exciton-phonon coupling

Singlet fission (SF) is a dynamical process, at the early stage of which singlet excited state (Frenkel exciton state, FE) on a single molecule or delocalized on two or more molecules is converted to the double-triplet state (TT) virtually or really through the charge transfer exciton state (CT) as shown in Fig. 10 [23,24]. As explained in Sec. 4, conventional studies mostly investigate the transition probability from the initial state (FE) to the final state (TT) based on Fermi's Golden rule (Eq. (8a)), using the electronic coupling matrices (Eq. (9)), which depend on the relative molecular configuration in the crystal as shown in Sec. 4. On the other hand, the SF process has time-dependent population variations in plural states and is also affected by coherence variations between these states by including relaxation process due to the exciton-phonon (vibronic) coupling, which cannot be sufficiently described by Fermi's Golden rule. Therefore, recent several theoretical studies shed light on the importance of SF dynamics by using various non-adiabatic exciton dynamics approaches including (non-)Markovian density matrix method [29,30,102,123,161,163,164,186], multiconfigurational time-dependent Hartree method [169,187], surface hopping method [97,188,189], and symmetrical quasiclassical simulation method [163,164,190]. Owing to intensive developments of experimental techniques, such ultrafast vibrational (non-adiabatic) dynamics are now targets of experimental studies. Therefore, dynamical description of singlet fission is inevitable for comprehensive understanding of the mechanism and constructing a concrete design guideline. In this section, we introduce the non-Markovian density matrix propagation scheme since it can provide an easy tool to analyze the SF dynamics based on the relaxation factors, which are understood by the exciton-phonon coupling spectral density obtained from the vibronic coupling analysis shown in Sec. 5.2.

Let us consider the total Hamiltonian of the molecular dimer:

$$H = H_{\text{ex}} + H_{\text{ph}}^{\text{H}} + H_{\text{ph}}^{\text{P}} + H_{\text{ex-ph}}^{\text{H}} + H_{\text{ex-ph}}^{\text{P}}, \quad (21)$$

where H_{ex} is the exciton Hamiltonian, and H_{ph}^{X} is the phonon Hamiltonian for the Holstein coupling (X = H) and for Peierls coupling (X = P). $H_{\text{ex-ph}}^{\text{X}}$ indicates the exciton-phonon coupling Hamiltonian for the Holstein coupling (X = H) and the Peierls coupling (X = P). The exciton Hamiltonian is represented as

$$H_{\text{ex}} = \sum_{m=1}^5 H_{mm} |m\rangle\langle m| + \sum_{n < m}^N H_{mn} (|m\rangle\langle n| + |n\rangle\langle m|), \quad (22)$$

where $\{|m\rangle\}$ indicates the exciton basis: $\{|1\rangle, |2\rangle, |3\rangle, |4\rangle, |5\rangle\} = \{|S_1S_0\rangle, |S_0S_1\rangle, |CA\rangle, |AC\rangle, |TT\rangle\}$ (see Eqs. (6a–f), where for CA, AC, and TT, super/sub-scripts are omitted for simplicity).

Using Eq. (35) and the five exciton basis $\{|S_1S_0\rangle, |S_0S_1\rangle, |CA\rangle, |AC\rangle, |TT\rangle\}$, the exciton Hamiltonian of the dimer is approximately expressed by [123]

$$H_{\text{ex}} = \begin{pmatrix} |S_1S_0\rangle & |S_0S_1\rangle & |CA\rangle & |AC\rangle & |TT\rangle \\ \begin{matrix} E_{S_1S_0} & V_{\text{ex}} & V_{ll} & -V_{hh} & V_{2e} \\ V_{\text{ex}} & E_{S_0S_1} & -V_{hh} & V_{ll} & V_{2e} \\ V_{ll} & -V_{hh} & E_{\text{CT}} & V_{\text{CT}} & \sqrt{3/2}V_{lh} \\ -V_{hh} & V_{ll} & V_{\text{CT}} & E_{\text{CT}} & \sqrt{3/2}V_{hl} \\ V_{2e} & V_{2e} & \sqrt{3/2}V_{lh} & \sqrt{3/2}V_{hl} & E_{\text{TT}} \end{matrix} \end{pmatrix} \quad (23)$$

Here, V_{hh} and V_{ll} indicate the one-electron couplings and between the HOMOs ($\langle h_A | F | h_B \rangle$) and between the LUMOs ($\langle l_A | F | l_B \rangle$) of the two molecules, respectively (see Eqs. (9a–d)). In these matrix elements, the two-electron couplings are approximately ignored due to the relative small amplitudes, though the direct contribution V_{2e} is reported to play an important role in some recent studies [120,123]. The excitonic couplings between S_1S_0 and S_0S_1 and between CA and AC are represented by V_{ex} and V_{CT} , respectively.

On the other hand, we can define the symmetry-adapted basis $\{|S_{\pm}\rangle, |CT_{\pm}\rangle, |TT\rangle\}$, the first two bases are

$$|S_{\pm}\rangle = (\sqrt{2})^{-1} (|S_1 S_0\rangle \pm |S_0 S_1\rangle), \quad (24a)$$

$$|CT_{\pm}\rangle = (\sqrt{2})^{-1} (|CA\rangle \pm |AC\rangle), \quad (24b)$$

then we obtain the electronic Hamiltonian of the dimer represented in symmetry-adapted basis:

[23,97,168]

$$H_{\text{ex}} = \begin{pmatrix} |S_+\rangle & |CT_+\rangle & |TT\rangle & |CT_-\rangle & |S_-\rangle \\ E_{S_1 S_0} + V_{\text{ex}} & V_{ll} - V_{hh} & \sqrt{2}V_{2e} & 0 & 0 \\ V_{ll} - V_{hh} & E_{CT} + V_{CT} & (\sqrt{3}/2)(V_{hl} + V_{lh}) & 0 & 0 \\ \sqrt{2}V_{2e} & (\sqrt{3}/2)(V_{hl} + V_{lh}) & E_{TT} & (\sqrt{3}/2)(V_{hl} - V_{lh}) & 0 \\ 0 & 0 & (\sqrt{3}/2)(V_{hl} - V_{lh}) & E_{CT} - V_{CT} & V_{ll} + V_{hh} \\ 0 & 0 & 0 & V_{ll} + V_{hh} & E_{S_0 S_1} - V_{\text{ex}} \end{pmatrix} \quad (25)$$

The second-order time-convolutionless (TCL) Master equation is expressed by

$$\frac{d}{dt}\rho_{\text{ex}}(t) = -\frac{i}{\hbar}[H_{\text{ex}}, \rho_{\text{ex}}(t)] + \sum_{X=H,P} \sum_{\omega, m} \gamma_X(\omega, t) \left(A_m^X(\omega) \rho_{\text{ex}}(t) A_m^X(\omega)^\dagger - \frac{1}{2} \{ A_m^X(\omega)^\dagger A_m^X(\omega), \rho_{\text{ex}}(t) \} \right) \quad (26)$$

Here [191],

$$A_m^X(\omega) = \mathring{\mathbf{a}}_{E_a - E_b = \omega} |Y_a\rangle \langle Y_a| A_m^X |Y_b\rangle \langle Y_b|, \quad (27a)$$

where $\{|Y_a\rangle, E_a\}$ are an eigenstate ($|Y_a\rangle (= \mathring{\mathbf{a}}_m C_{ma} |m\rangle)$) and an eigenvalue of H_{ex} :

$H_{\text{ex}}|Y_a\rangle = E_a|Y_a\rangle$. The time-dependent relaxation rate $g_X(\omega, t)$ is

$$g_x(w, t) = 2 \int_0^\infty dn J_x(n) \left([1 + n(n)] \frac{\sin(w - n)t}{w - n} + n(n) \frac{\sin(w + n)t}{w + n} \right). \quad (27b)$$

Here, $n(n)$ is the Bose-Einstein distribution, and $J_x(n)$ is spectral density of vibrational mode X (= H (Holstein) and P (Peierls)). For example, $J_x(n)$ is approximated by the Ohmic function with Lorentz-Drude cut off [30,102] with the reorganization energy λ_x and the cutoff frequency Ω_x of mode X. Note here that the bath spectral density can take other functional forms.

5.4. Applications of exciton dynamics, including exciton-phonon coupling

Let us consider singlet fission dynamics in a pentacene dimer as a model of typical SF systems [97]. Wang et al. evaluated SF dynamics of a pentacene dimer by using the self-consistent fewest switches surface hopping approach [97]. As seen from Fig. 28A, high E_{CT} is found to be good for achieving a high net triplet yield for SF (due to avoiding the creation of excimer (strongly mixed FE–CT) states with lower energy than the TT state), the feature of which corresponds to a relatively large distance between the two pentacene molecules in the dimer. From the 2D map of SF time constant τ (Fig. 28C), fast SF is observed in several localized regions indicated by I, II, III, and IV, all of which correspond to configurations slipped along the transverse direction for approximately one ring (see Fig. 28E). From Eq. (25), $V_{hl} = -V_{lh}$ and $V_{hh} = V_{ll}$ are found to maximize the SF transition rate for the same ΔE_{CT} . They found that the electronic couplings V_{hh} (t_{HH}) and V_{ll} (t_{LL}) have large distribution with the same sign along a wide region in the T direction. In contrast, V_{hl} (t_{HL}) and V_{lh} (t_{LH}) have similar amplitude distribution with opposite sign except for the middle T region on the 2D map. The overlap region between these two regions satisfying the distribution maximizing Eq. (11) correspond to the fast SF (small SF time constant) region. The relationship between the SF time constant and CT character are shown in Fig. 28D. A large CT character tends to have a smaller SF time constant. However, since the data points spread out of the behavior, there is no clear relationship between the SF time constant and the CT character. From Eq. (25), which uses symmetry adapted basis states $\{|S_+\rangle, |S_-\rangle, |CT_+\rangle, |CT_-\rangle, |TT\rangle\}$, there are two ways for generating triplets from the initial singlet

exciton state, (1) $S_+ \rightarrow CT_+ \rightarrow TT$ (the symmetric channel) and (2) $S_- \rightarrow CT_- \rightarrow TT$ (the asymmetric channel). The efficiency of (1) depends on $|V_{hh} - V_{ll}|$ and $|V_{hl} + V_{lh}|$, while (2) depends on $|V_{hh} + V_{ll}|$ and $|V_{hl} - V_{lh}|$. From these considerations, they found that singlet fission is inefficient for (1), while fast for (2), when low temperature, 0 K. Interestingly, this kind of destructive interference effect is also found to collapse with increasing temperature, and the key features are described by the 2D map of each matrix element in Eq. (23).

Next, considering again Eqs. (26), (27a) and (27b), Peierls coupling effect is discussed. This effect was usually considered to be small because of the cut frequency $\omega_p \ll \omega_H$ and the reorganization energy $\lambda_p \ll \lambda_H$. However, recent experimental and theoretical studies pointed out that Peierls coupling significantly affects the singlet fission dynamics and sometimes opens an additional singlet fission channel [27,110,114,123,160,167,180]. Using the electronic coupling Hamiltonian (Eq. (25)), Renaud et al. have elucidated the variation in population of TT at $t = 100\text{ps}$ ($T = 300\text{K}$) without and with Peierls coupling for sequential (Seq) and superexchange (SpX) regions (Fig. 29) [123]. As seen from Fig. 29a, fast singlet fission occurs in the Seq region, but also does in a limited area in SpX region in the case of $0 < E_{CT} - E_{TT} < 0.5\text{ eV}$. This feature is in agreement with the fast indirect pathway predicted by Michl et al. (see Eq. (10)) when ΔE_{CT} is less than $\sim 0.5\text{ eV}$. In contrast, as seen from Fig. 29b, considering Peierls coupling, fast singlet fission occurs regardless of CT energy ΔE_{CT} as long as $E_{FE} - E_{TT} < \sim 0.25\text{ eV}$. This is understood by the fact that the direct pathway transition probability is increased due to Peierls coupling (which causes the direct transition from FE to TT by emitting a phonon concerning the related vibration mode, see Eqs. (27a and b)). In realistic systems, it has been demonstrated that the modulations of the two-electron coupling induced by the intermolecular vibrational modes of a molecular crystal can significantly affect the singlet fission rate [123]. This is realized under the following conditions: (i) $|E_{FE} - E_{TT}|$ should not be too large as compared to the characteristic frequency of the intermolecular vibrational modes, (ii) E_{CT} should be relatively high and the one-electron couplings relatively weak in order to limit the

influence of the indirect mechanism, and (iii) the two-electron coupling should show large variations around the equilibrium position of the molecular dimer. Indeed, several perylenediimide (PDI) derivatives are shown to satisfy all of these conditions, and thus the singlet fission rates calculated for these dimers are found to be sensitive to the Peierls coupling: a significant increase of the singlet fission rate, of about 1 order of magnitude, was obtained for the selected PDI derivatives [123]. These results contribute to rationalizing the sub-picosecond singlet fission rate measured for the selected molecular crystals despite the ΔE_{CT} [123].

The singlet fission rate is also found to depend on the spectral density of exciton-phonon coupling [30,123,163,164,166–168]. The energy gap between the adiabatic electronic states and peak frequency of the vibronic coupling significantly affect the amplitude of relaxation rate as shown in Eq. (27b). Furthermore, the distribution shape of spectral density also has an impact on the relaxation rate, e.g., it is found that the effect of the Peierls coupling, which usually has a peak at low frequency, is almost negligible for a super-Ohmic spectral density, while it remains important in the case of a sub-Ohmic density because of their different high-frequency cutoff [123]. In the sub-Ohmic case, the spectral density still has large distribution at the frequency corresponding to energy gap between E_{FE} and E_{TT} , so that the Markovian relaxation rate induced by the Peierls coupling is predicted to be non-negligible and the SF dynamics is expected to be improved by the Peierls coupling concerning intermolecular vibration modes. In contrast, in the case of super-Ohmic spectral density which is negligible at the frequency corresponding to $E_{FE} - E_{TT}$, the Peierls coupling effect is predicted to be negligible. As a result, the high frequency cutoff of the spectral density significantly affects the singlet fission dynamics. However, even in the case of super-Ohmic, the Peierls coupling effect can be important if the difference between E_{FE} and E_{TT} is small. In addition, in the case of fast SF dynamics, non-Markov effect, which is expected to enhance the SF rate in the early time region even in the spectral peaks in the lower frequency region than the energy gap $|E_{FE} - E_{TT}|$, may contribute to increasing the SF efficiency.

Fujihashi and co-workers presented significant effects of the characteristic frequency of Holstein and Peierls coupling, and of relative amplitudes between the direct electronic coupling and vibronic coupling (reorganization energy) [166,167]. They used a two state (S_1S_0 and TT) model, thus the direct electronic coupling may be understood as an effective one, Eq. (8a). Calculated singlet population evolution shows interesting patterns for each characteristic frequency region, see Fig. 30. In Fig. 30, $\hbar\omega_{diag}$ and $\hbar\omega_{o.d.}$ are the characteristic frequencies of Holstein and Peierls couplings. When low $\hbar\omega_{o.d.} < 50$ meV, singlet fission dynamics is mainly promoted by only Holstein coupling around $\hbar\omega_{diag} = 80$ meV, which causes the resonance between S_1S_0 and TT in this calculation condition. Peierls coupling in this $\hbar\omega_{o.d.}$ region does not do anything due to small reorganization energy in the off-resonant region. In the high-frequency $\hbar\omega_{o.d.} > 80$ meV region, singlet fission is mainly facilitated by Peierls coupling. In the intermediate region, two channels induced by Holstein and Peierls couplings are shown to coexist. It was also found that such interesting effect of vibronic coupling is not seen when the direct (or maybe effective) electronic coupling is large [167]. This suggests that roles of electronic coupling and vibronic coupling are not independent but are related with each other. They also provided a viewpoint of energy level matching condition in vibronic eigenstates, by virtue of a wavefunction ansatz for an electron–phonon problem [166].

5.5. State-dependent spectral density effects on singlet fission dynamics

We here consider the effect of state-dependent spectral density on singlet fission dynamics by using artificial herringbone pentacene dimer model [168], whose geometry is obtained from an experimental study [192]. The electronic coupling matrix elements for the herringbone pentacene dimer model is given in Table 3. In the case of the realistic pentacene dimer model, the amplitudes of V_{ex} , V_{CT} and V_{2e} are assumed to be zero [23,24,30].

In order to investigate the exciton relaxation between the adiabatic exciton states $\{|Y_a\rangle\}$, the relative relaxation factor (RRF) [168,193,194] is introduced. In the case of Holstein coupling at the Markovian limit, it is expressed by

$$DG(a \rightarrow b) = \sum_m |C_{ma}|^2 |C_{mb}|^2 [g_m(w_a - w_b) - g_m(w_b - w_a)], \quad (28)$$

where the first and second terms in square bracket on the right-hand side indicate the relaxation rate from state α to β and that from state β to α , respectively. The positive (negative) value of

$DG(a \rightarrow b)$ indicates the exciton relaxation from state α to β (from state β to α), and its amplitude indicates the magnitude of the net relaxation rate between states α and β . The important point in

Eq. (28) is that $DG(a \rightarrow b)$ has a large amplitude when both the adiabatic exciton states α and β

include common diabatic exciton $\{|m\rangle\}$ configurations (non-zero $|C_{ma}|^2 |C_{mb}|^2$), and that

$DG(a \rightarrow b)$ amplitude depends on the relative energy differences in the adiabatic exciton states

(related to $[g_m(w_a - w_b) - g_m(w_b - w_a)]$) and the electronic couplings (related to $|C_{ma}|^2 |C_{mb}|^2$).

Using the RRFs analysis, we can clarify the origin of the exciton relaxation rate (SF rate) and TT yield.

The SF dynamics is investigated for the pentacene dimer model with realistic parameters (Table 3). The initial state population is assumed to 1 in $|S_1 S_0\rangle$ since the feature of SF dynamics (SF rate and TT yield) little depends on the kinds of initial state superpositions [161]. The results are shown in Fig. 31a. The adiabatic exciton states $\{|a\rangle\} = \{|S_+\rangle, |S_-\rangle, |CT_+\rangle, |CT_-\rangle, |TT\rangle\}$, which are obtained by diagonalizing the electronic coupling matrix, represent superposition states of

$$|S_+\rangle = (|S_1 S_0\rangle + |S_0 S_1\rangle) / \sqrt{2}, \quad |S_-\rangle = (|S_1 S_0\rangle - |S_0 S_1\rangle) / \sqrt{2}, \quad |CT_+\rangle = (|CA\rangle + |AC\rangle) / \sqrt{2},$$

$$|CT_-\rangle = (|CA\rangle - |AC\rangle) / \sqrt{2} \text{ and } |TT\rangle. \text{ As shown in Eqs. (28) and (27b), the exciton relaxation}$$

between the adiabatic exciton states depends on the relaxation rate $g_m(w)$, which is a function of

the spectral density $J_m(w)$ concerned with the exciton-phonon (vibronic) coupling. Although this

spectral density depends on the diabatic exciton state $\{|m\rangle\}$ in principle, such state-dependence has

not been investigated in most previous studies [30,123]. We here examine the effects of such dependence on the SF dynamics by using different spectral density (J_m , W_m) for each diabatic state $\{|m\rangle\}$, where we employ a reasonable approximation that the same parameters are employed for Frenkel exciton states $|FE\rangle$ ($= |S_1S_0\rangle$ or $|S_0S_1\rangle$) and for charge transfer states $|CT\rangle$ ($= |CA\rangle$ or $|AC\rangle$), respectively. As seen from our previous study [162], the feature of vibronic couplings are clarified for a tetracene dimer model using vibronic coupling density analysis, and it is found (i) that FE state exhibits large vibronic coupling peaks in the high frequency region ($1200 - 1700 \text{ cm}^{-1}$ ($150 - 210 \text{ meV}$)), for example, carbon(C)-carbon(C) stretching mode, while TT state exhibits almost the same spectral shape with these peaks, the amplitudes of which are twice those of FE state, and (ii) that CT state exhibits larger peaks in the low frequency region ($60 - 800 \text{ cm}^{-1}$ ($7 - 100 \text{ meV}$)). These larger spectral peaks in TT and CT states than in FE state are predicted to significantly affect the SF dynamics through Eq. (27b). To clarify the effects of state-dependent spectral density on the SF dynamics, we examine three model spectral densities, (J_m , W_m) = (50, 180) meV for FE, (1290, 98) meV for CT, and (200, 180) meV for TT as shown in Fig. 32, where the spectral density takes a peak (reorganization energy λ) at the cutoff frequency Ω . Fig. 31b shows the diabatic exciton population dynamics for the realistic pentacene dimer model with these state-dependent spectral densities. In comparison with the identical spectral density case (J , W) = (50, 180) meV for all the diabatic states (Fig. 31a), in the case of the state-dependent spectral density, the SF time constant τ is found to be remarkably reduced, while the TT yield is unchanged. As seen from the adiabatic state energy diagrams with RRFs for these cases (Fig. 31a, b), there are no differences in the coefficients of all the configurations in each adiabatic state between these two spectral density cases due to the same electronic coupling matrix, the feature of which indicates that the differences in RRFs originate in the state-dependent spectral densities. In the present case, we found the enhancement of RRFs between the adiabatic states including large CT and/or TT mixings due to the significantly larger reorganization energy in CT and TT states than in FE state [175].

In order to elucidate the effect of the state-dependent spectral density on the SF dynamics in a wide range of the energy offsets, $E(\text{CT}) - E(\text{TT})$ and $E(\text{FE}) - E(\text{TT})$, we compare the TT yield maps at 50 fs, 100 fs, 300 fs and 1000 fs in the identical spectral density case (Fig. 33a–d) with those in the state-dependent spectral density case (Fig. 33e–h). As compared to the identical spectral density case, the rapid evolution of large TT population is observed in the state-dependent spectral density case, in particular, in $\text{Spx}^+(\text{TT})$ region with high $|E(\text{CT}) - E(\text{TT})|$ and $|E(\text{CT}) - E(\text{FE})|$ values. Namely, in such $\text{Spx}^+(\text{TT})$ region, we can realize the efficient SF dynamics with both large SF rates and high TT yields if the reorganization energy of CT state is significant even in the low frequency region. In summary, significant state dependence of spectral density has a possibility of affecting the RRF $DG(a \rightarrow b)$ through the common diabatic exciton state mixings in the concerned adiabatic states (α and β) and thus of changing the SF rate with keeping the TT yield. Such large vibronic coupling at low frequency region of CT state is often observed for dimers of acenes and other conjugated organic molecules [162], so that the optimal tuning of energy level matching of the monomer, CT energy offset and electronic coupling is desired for further improving the SF efficiency for such systems.

In summary, for rational design of highly-efficient SF materials, the relationships between the monomer structures, crystal packing and the electronic/vibronic couplings, which are not mutually independent but correlated, have to be clarified in detail for a variety of SF candidate monomers screened by the diradical character based design guideline [49,50,53]. In addition, we need to tackle the following issues: (i) CT offset energy tuning by changing the ionization potential and electron affinity of monomer and intermolecular interaction between positively- and negatively-charged monomers as well as by designing the crystalline environment of polarizable molecules, and (ii) investigation of Peierls coupling effects, e.g., intermolecular vibronic coupling effects [123], where consideration of non-Markovian nature [186] in the exciton relaxation dynamics, i.e., the phonon-bath engineering, would be important for tuning the SF dynamics. The present quantum master equation approach with RRF analysis will be useful for such investigations.

6. Summary and perspectives

In this review, we introduce the theoretical and experimental studies on singlet fission from the viewpoint of material design. As shown in Fig. 2, the studies of singlet fission are divided into three steps ranging from the molecular level to the aggregate level: (i) single molecular design (energy level matching), (ii) molecular configuration design (electronic coupling control), and (iii) exciton dynamics design (vibronic coupling design or phonon bath engineering). Note that these steps are mutually correlated and the design guidelines in each step often affect each other. For example, the energy matching conditions at single molecule level may not be necessarily appropriate at the aggregate level since in the latter level, intermolecular interaction, i.e., electronic coupling, often significantly changes the relative excited energy levels and thus affects the energy matching feature at the single molecule level. In this review, we show some guidelines at step (i) (diradical character based design principle: relatively smaller diradical character and much smaller tetraradical character) and (ii) (slip-stack molecular configuration suitable for increasing the effective electronic coupling), and introduce recent studies on the vibronic coupling and singlet fission dynamics at step (iii). Interestingly, as seen from the analysis at step (iii), the molecular configurations with the maximum effective electronic couplings obtained at step (ii) do not necessarily lead to large double triplet (TT) yield [97,151]. The relaxation factor between adiabatic states depends on the common configurations (S_1S_0 , S_0S_1 , CA, AC, TT) included in the corresponding adiabatic states, which are determined by the diagonal and off diagonal parts of the electronic Hamiltonian, i.e., by the monomer energy levels and molecular configurations. In addition, the relaxation factor also depends on the vibronic couplings (Holstein and Peierls couplings concerning the intra- and intermolecular vibration), which are determined in principle by the electronic structures and molecular/aggregate structures. From these considerations, advances in analysis of the mechanism as well as in rational design of efficient singlet fission require the comprehensive investigation of singlet fission rate and double triplet yield, which are indistinguishably correlated to the three research steps including energetic and dynamic aspects of singlet fission. Namely, further detailed investigations at each step

as well as proposing the unified guidelines over these three steps are indispensable for building practical design guidelines for efficient singlet fission materials. Further developments of theoretical calculation methods applicable to a variety of realistic molecules as well as to their various molecular configurations are necessary for this end.

Note that we have not discussed in detail how a correlated triplet pair (TT) turns into separated triplet excitons ($T + T$). This issue remains unclear and is under discussion. Separation of a correlated triplet pair could not be independent from an efficient generation of a correlated triplet pair, but closely related to each other. Recent experimental studies have illuminated this issue [41,195–197], and further developments of both experimental and theoretical studies are desired to reveal the mechanism.

The opposite process to singlet fission, triplet–triplet annihilation (TTA), has attracted much attention from the viewpoint of a promising upconversion approach due to its utilization of low excitation power density like solar light, readily tunable excitation/emission wavelength, strong absorption of the excitation light, and high upconversion quantum yields [198,199]. Although the mechanism of upconversion by TTA is yet to be clarified in detail, various types of TTA based upconversion systems are extensively investigated. The clarification of the singlet fission mechanism is expected to contribute to revealing TTA process. On the other hand, such singlet – triplet interconversion processes are also expected to speed up the exciton migration and to enhance the lifetime in organic crystals by repeated fission and recombination of a singlet exciton and triplet excitons, respectively. The study by Huang and colleagues have shown this unique phenomena that is not seen in usual excitons [197]. It may be expected that the efficiency of energy transport over micrometer length scales is improved in organic crystals through the design of materials with proper balance between SF and TTA. Thus, the detailed mechanisms and control schemes for SF as well as for TTA must be clarified based on the comprehensive research steps as shown in this review.

Further close cooperation between theoretical and experimental researchers over these three research steps is necessary for tackling these issues.

Although the present intensive exploration of singlet fission has been historically started after a proposal of photovoltaic cell application, a generation of an entangled pair of high-spin (triplet) excitons may show another potential for future applications in the field of quantum computer and spintronics [35,200]. For the former application, a long entanglement lifetime is critically important. Entanglement lifetime of correlated spins generated from singlet fission can be longer than a nanosecond [196], though the possibility of its extension to micro- and milliseconds has not been demonstrated yet. Further modifications of chemical moieties, and probably physical environment (temperature, pressure, etc.), may enhance the entanglement lifetime drastically [201,202]. Such a variety of fascinating research fields bringing out the outstanding potentials of singlet fission is vastly expanded before our eyes.

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Biography

Soichi Ito received his Ph.D (2017) degree in Science from Osaka University under the supervision of Prof. Masayoshi Nakano (2011–2017). He was a Research Fellowship for Young Scientists of Japan Society for the Promotion of Science (2014–2017). He has been interested in open-shell singlet molecules and singlet fission, especially of their theoretical molecular design. He is currently the assistant professor in Prof. Masahiro Ehara's group in Institute for Molecular Science in Japan since 2017.



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Table 1. CT-Mediated energy perturbation to FE and TT states, Coulomb and CT-mediated excitonic coupling, and FE-TT coupling of tetracene dimers **Tc1-(*m,n*)** and **Tc2-(*m,n*)** in meV evaluated at LC-BLYP/6-31G** level of theory.

		Tc1-(<i>m,n</i>)			Tc2-(<i>m,n</i>)		
		1	<i>n</i> 2	3	1	<i>n</i> 2	3
		ΔE_{FE}					
<i>m</i>	1	-566.9			-514.7		
	2	-29.3	-284.2		-12.8	-218.3	
	3	-374.9	-4.7	-165.4	-255.6	-2.9	-120.0
		ΔE_{TT}					
<i>m</i>	1	-593.4			0.0		
	2	-58.8	-411.7		-420.5	0.0	
	3	-169.0	-10.0	-188.8	-38.0	-212.2	0.0
		J_{Coul}					
<i>m</i>	1	23.4			1.4		
	2	14.8	9.0		1.8	0.8	
	3	10.4	6.3	4.4	3.3	0.9	0.5
		J_{CT}					
<i>m</i>	1	-199.9			512.0		
	2	16.5	255.3		8.1	218.3	
	3	335.7	3.2	148.9	250.7	2.3	119.6
		V_{FE-TT}					
<i>m</i>	1	427.0			0.0		
	2	3.9	78.2		-31.9	0.0	
	3	96.4	2.8	40.0	-9.7	-8.4	0.0
		$(E_{TT} - E_{FE})^{eff} = (E_{TT} + \Delta E_{TT}) - (E_{FE} + \Delta E_{FE} - J_{Coul} + J_{CT})$					
<i>m</i>	1	40.1			918.1		
	2	-108.1	26.9		-507.7	327.3	
	3	441.9	-105.8	19.8	361.6	-316.1	130.1

Ref. [116] - Published by The Royal Society of Chemistry.

Table 2. Electronic couplings^a [meV] calculated at RM06/cc-pVDZ Level of theory and experimental SF time constant τ_{SF} [ps].

Molecule	Direct-overlap coupling			Bridge-mediated coupling			$ V_{\text{SF}} $	τ_{SF}^b
	F_{H1H2}	F_{H1L2}	F_{L1L2}	F_{H1H2}	F_{H1L2}	F_{L1L2}		
<i>o</i>-Pc	47	347	-117	120	97	82	169	0.5
<i>m</i>-Pc	6	-1	3	3	0	-1	0.01	63
<i>p</i>-Pc	1	-6	4	-129	-101	-81	10	2.7
Az-Pc	-2	6	-3	158	117	91	16	N/A
Nm-Pc	-2	2	0	40	26	18	1	N/A
Rad-Pc	15	-1	-9	444	329	244	138	N/A

^a F_{L1H2} is almost the same as F_{H1L2} due to the approximate symmetry of the molecular structures, and thus is not shown here.

^b Ref [111].

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Table 3. Electronic coupling matrix elements for the herringbone pentacene dimer model.

Electronic coupling matrix element [meV]	
$E(\text{FE})^a$	2120
$E(\text{CT})^a$	2320
$E(\text{TT})^a$	1720
V_{hh}^b	101.5
V_{ll}^b	-125.8
V_{hl}^b	104.6
V_{lh}^b	-121.7

^a See ref. [168].

^b Obtained from RB3LYP/cc-pVDZ calculations for the dimer model, whose geometry is extracted from an experimental study [192].

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Figures with captions

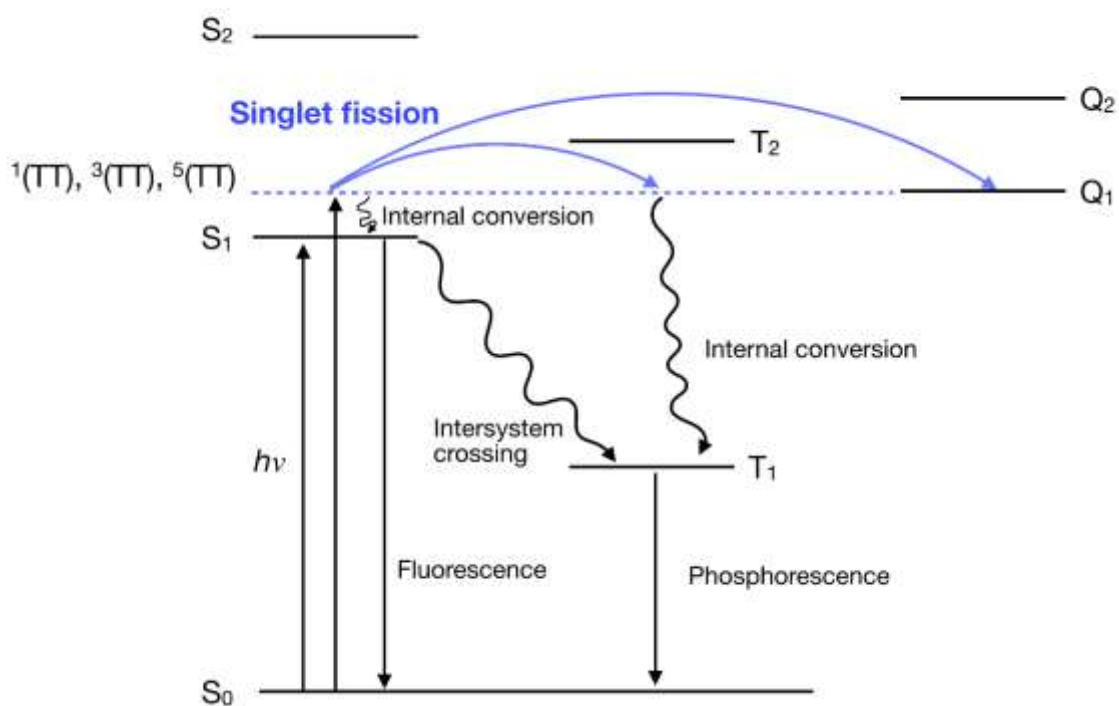


Fig. 1. Extended Jablonski diagram. Blue arrows indicate singlet fission process.

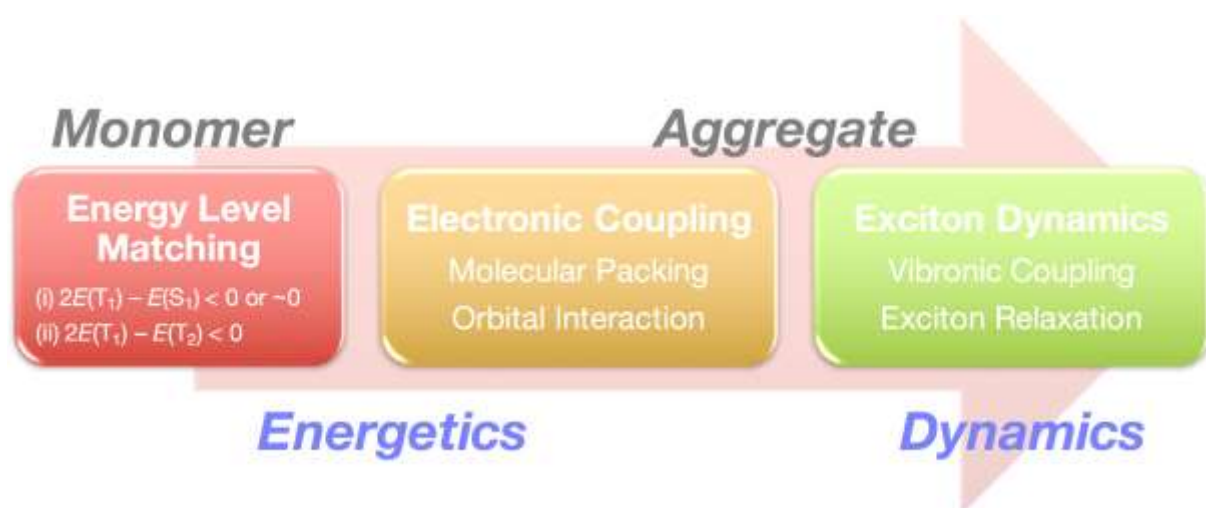


Fig. 2. Three steps of singlet fission study for molecular design.

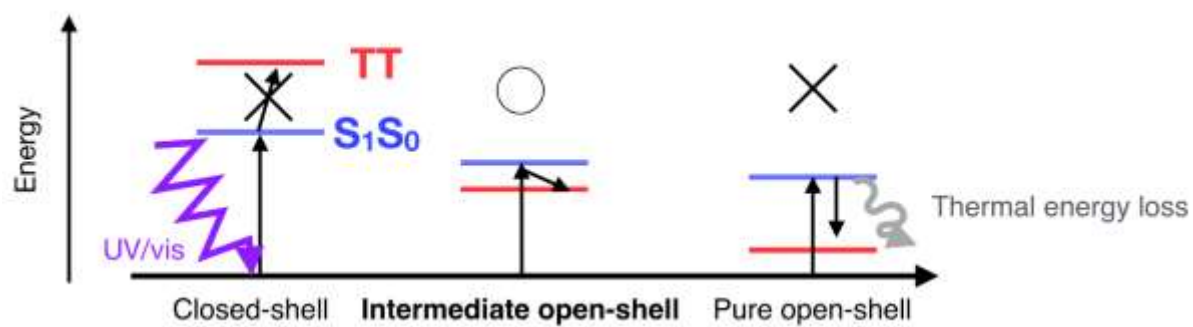


Fig. 3. Relationships between open-shell/closed-shell natures and excitation energies.

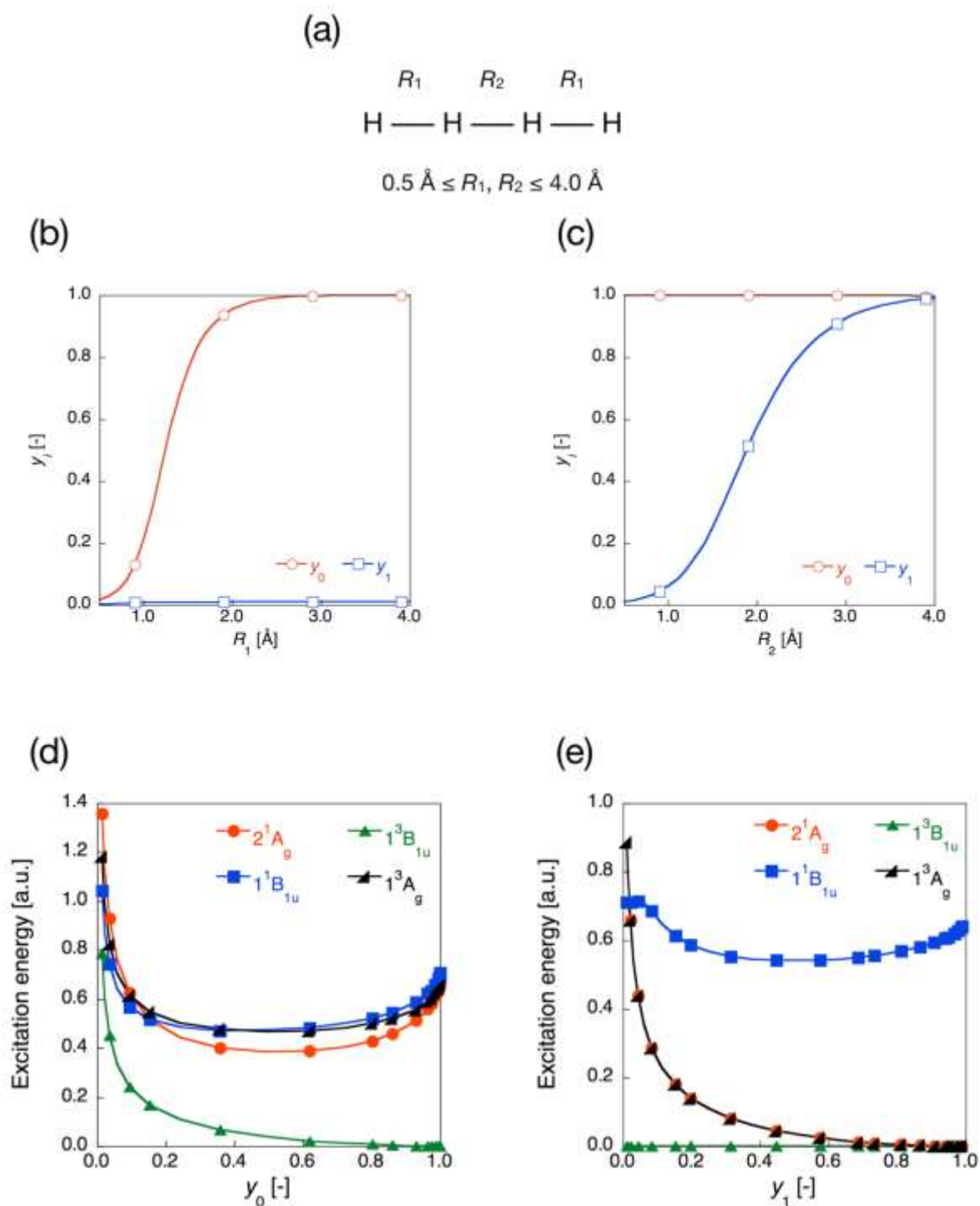


Fig. 4. Symmetric linear H_4 model (a) and variations of multiple diradical characters, y_0 and y_1 , as a function of R_1 with fixed $R_1 = 0.5 \text{ \AA}$ (b), and of R_2 with fixed $R_1 = 4.0 \text{ \AA}$ (c), variations of excitation energies of the 2^1A_g , 1^1B_u , 1^3B_u , and 1^3A_g states as a function of y_0 for $y_1 \sim 0$ (d) and of y_1 for $y_0 \sim 1$ (e). Reprinted with permission from Ref. [49] Copyright @ 2012 American Chemical Society.

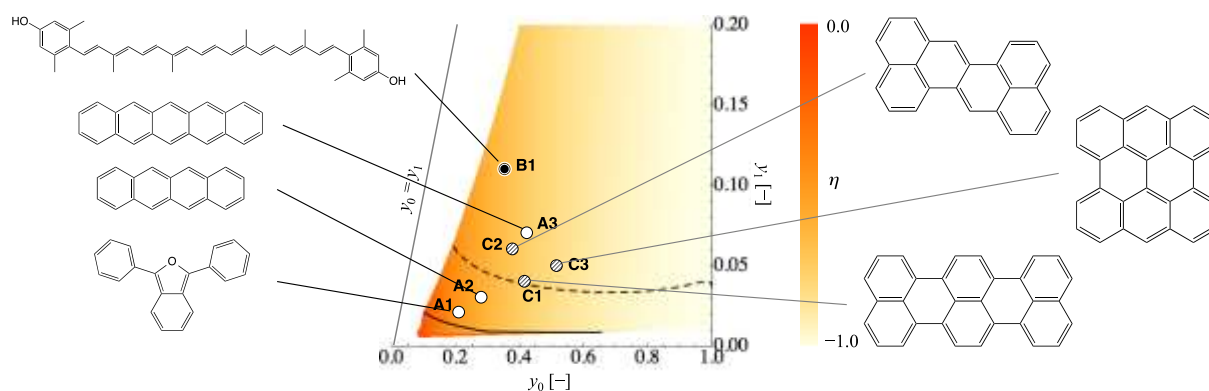


Fig. 5. Feasibility map of singlet fission on y_0 - y_1 plane based on the energy level matching conditions for the linear H_4 model (a) and several singlet fission molecules (**A1** (1,3-diphenylisobenzofuran), **A2**(tetracene), **A3**(pentacene), **B1**(zeaxanthin), **C1**(zethrene), **C2**(terrylene), **C3**(bisanthene)) (b). The density plot represents the energy efficiency with a function of $\eta = [2E(1^3B_u) - E(1^1B_u)/E(1^1B_u)]$, where the darker region has higher efficiency. The solid and dashed lines show $[E(2^1A_g) - E(1^1B_u)/E(1^1B_u)] = 0$ and -0.3 , respectively. The y_0 and y_1 are calculated at the PUHF/6-31G* level of approximation. Reprinted with permission from Ref. [49]. Copyright @ 2012 American Chemical Society.

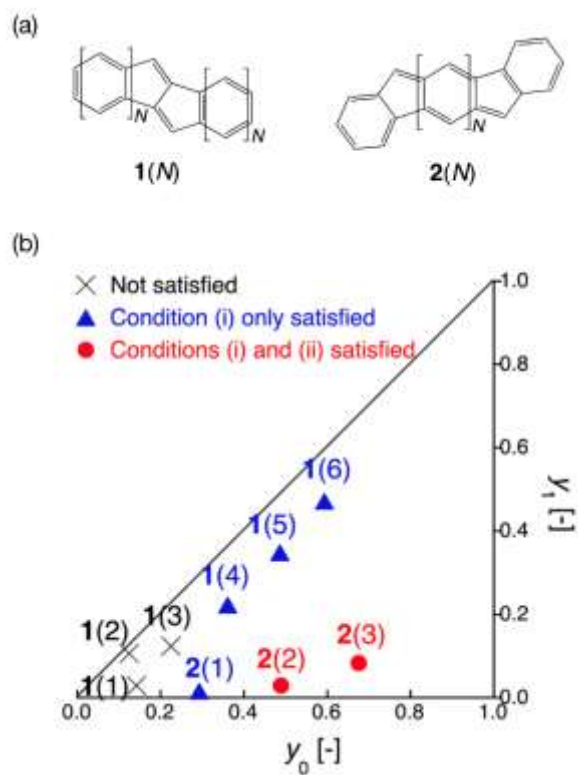


Fig. 6. Structures of condensed-ring molecules, **1(N)** and **2(N)**, where N indicates the number of fused rings (a), and diradical characters y_0 – y_1 map (b). Adapted with permission from Ref. [62].

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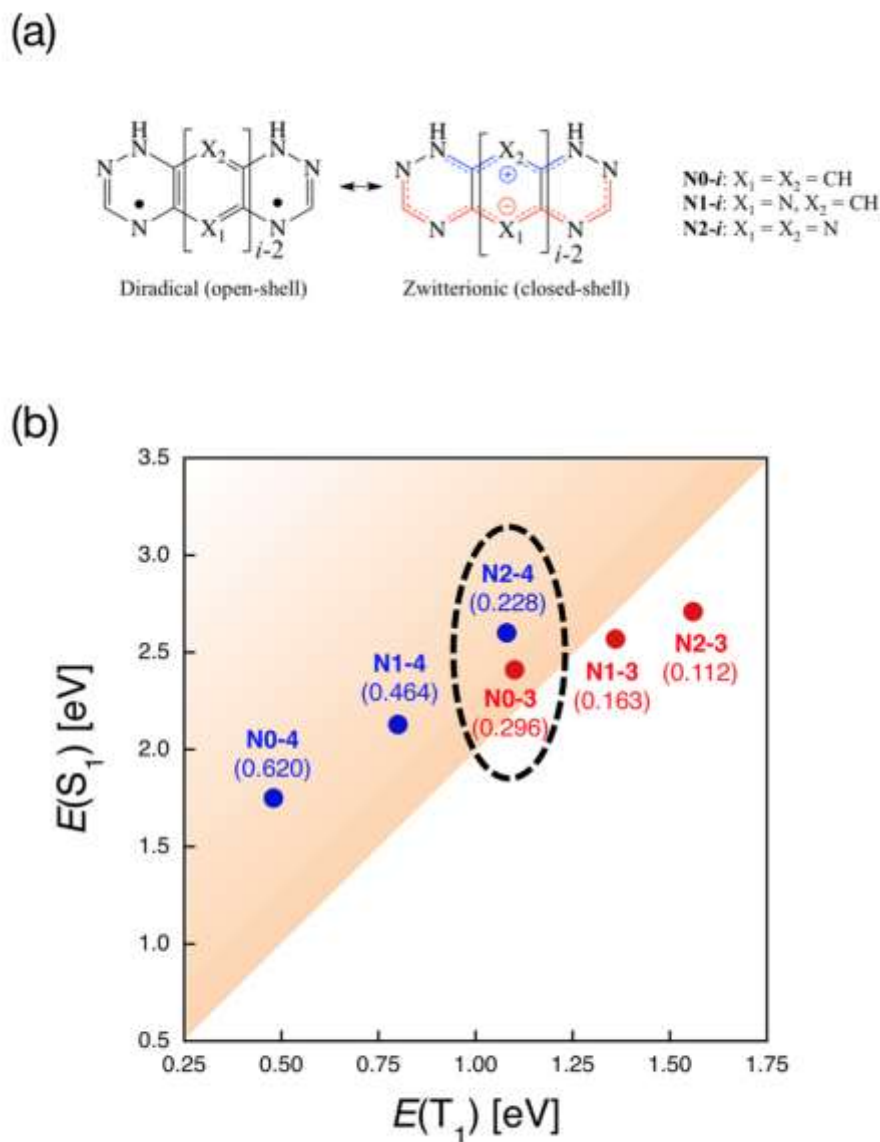


Fig. 7. (a) Heteroacene models **Nm-i** ($m = 0, 1, 2$; $i = 3, 4$). (b) Singlet [$E(S_1)$] and triplet [$E(T_1)$] excitation energies of heteroacene models. Red and blue circles represent three- ($i = 3$) and four- ($i = 4$) ring molecules, respectively. The diradical character y_0 values are shown in parentheses. In the colored region, the energy level matching condition (Eq. (2)) is satisfied, where darker (lighter) region shows the smaller (larger) energy-loss $|2E(T_1) - E(S_1)|$ region. Two promising candidates, **N0-3** and **N2-4**, which have $0.2 < y_0 < 0.4$, for efficient singlet fission are surrounded by a dotted circle.

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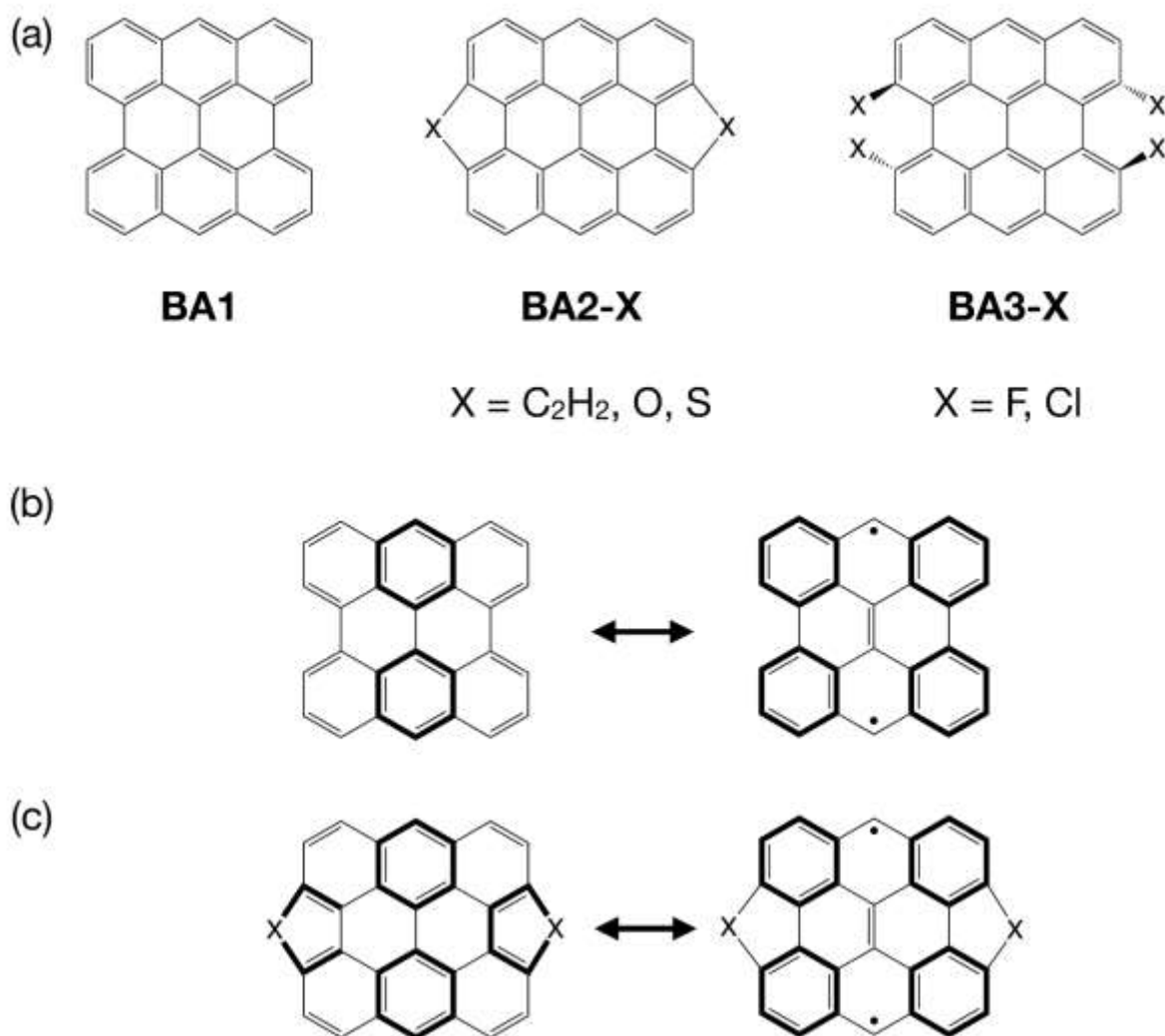


Fig. 8. Molecular structures of polycyclic aromatic hydrocarbons (a), and closed-shell and open-shell resonance structures of **BA1** (b), **BA2-X** (c) and **BA3-X** (d). In (b), (c) and (d), Clar's sextets are shown by bold solid lines. Reprinted with permission from Ref. [72]. Copyright @ 2016 American Chemical Society.

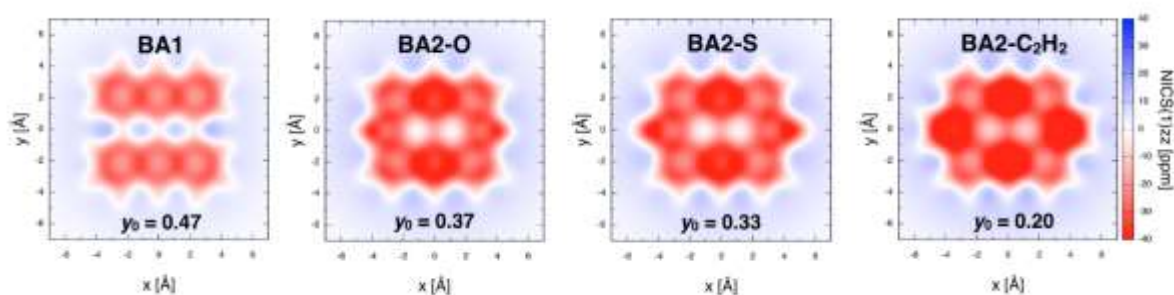


Fig. 9. NICS(1)_{zz} map for **1** and **2-X** calculated using the LC-BLYP/cc-pVDZ method. Red (negative) and blue (positive) distributions indicate aromatic and anti-aromatic regions, respectively. Reprinted with permission from Ref. [72]. Copyright @ 2016 American Chemical Society.

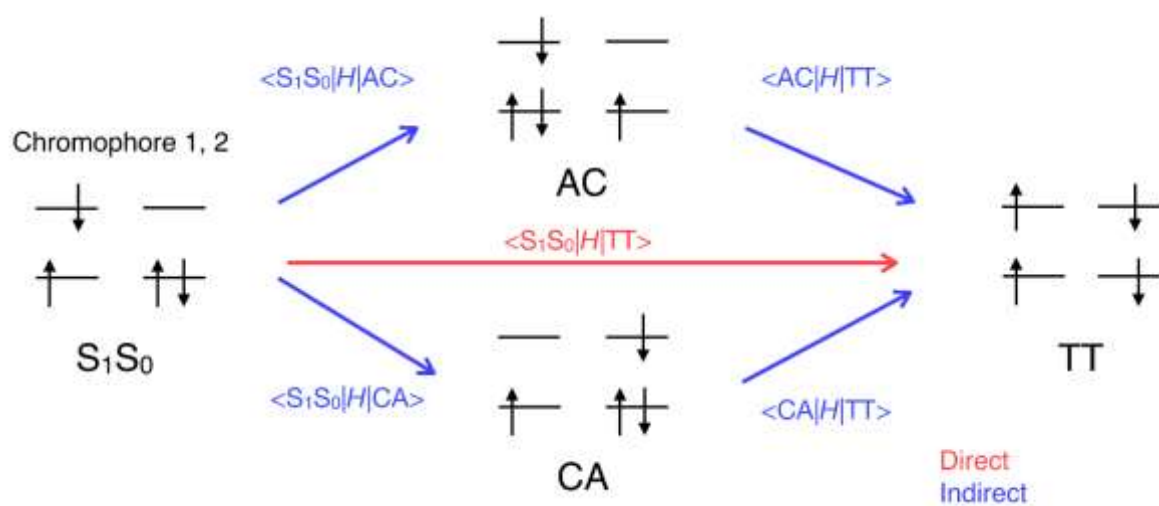


Fig. 10. Singlet fission mechanisms: direct (red arrow and matrix element) and in-direct pathways (blue arrows and matrix elements).

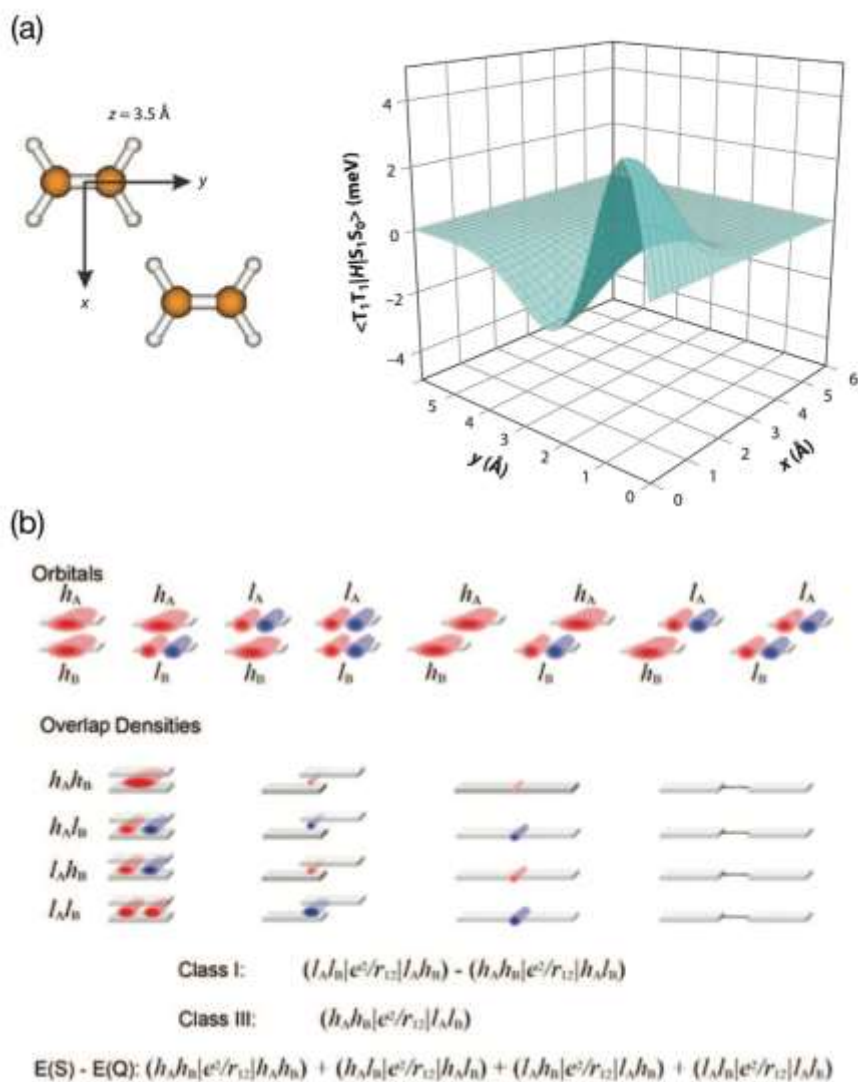


Fig. 11. Slip dependences of $\langle {}^1(\text{TT}) | H_{\text{el}} | S_1 S_0 \rangle$ of stacked ethylene dimers (at the HF/6-311G level of approximation) (a), and frontier orbitals (h_A , l_A , h_B , l_B) and their overlaps for different configurations (b). Reprinted with permission from the Annual Review of Physical Chemistry, Volume 64 (Ref. [24]) © 2013 by Annual Reviews, <http://www.annualreviews.org> (a), and Reprinted with permission from Ref. [23]. Copyright © 2010 American Chemical Society (b).

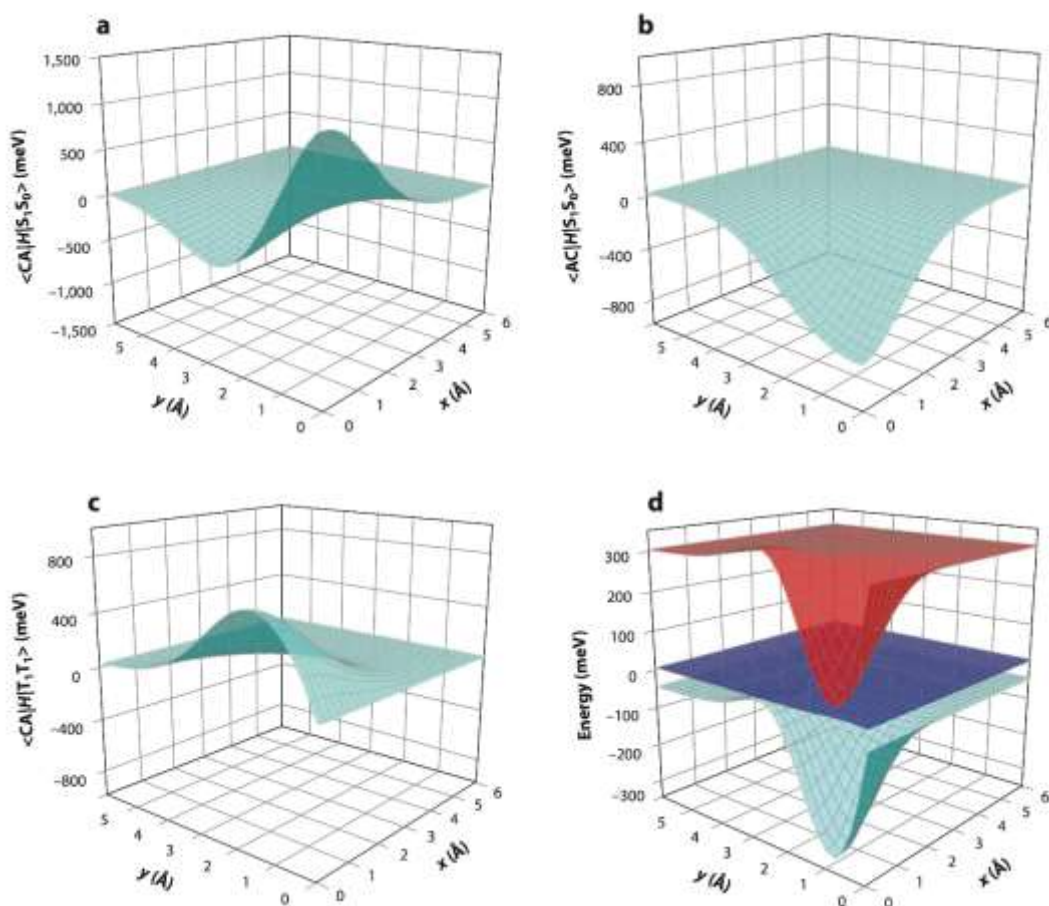


Fig. 12. Slip dependences of matrix elements in Eqs. (12) and (13) of stacked ethylene dimers (at the HF/6-311G level of approximation) [24]: $\langle {}^1CA | H_{el} | S_1 S_0 \rangle$ (a), $\langle {}^1AC | H_{el} | S_1 S_0 \rangle$ (b), $\langle {}^1(TT) | H_{el} | CA \rangle$ (c) and the second-order (indirect) contribution (red), the first-order (direct) contribution (blue), and their sum (pale blue) (50 kcal/mol is assumed for DE_{CT}) (d). Reproduced with permission of Annual Review of Physical Chemistry, Volume 64 (Ref. [24]) © 2013 by Annual Reviews, <http://www.annualreviews.org>.

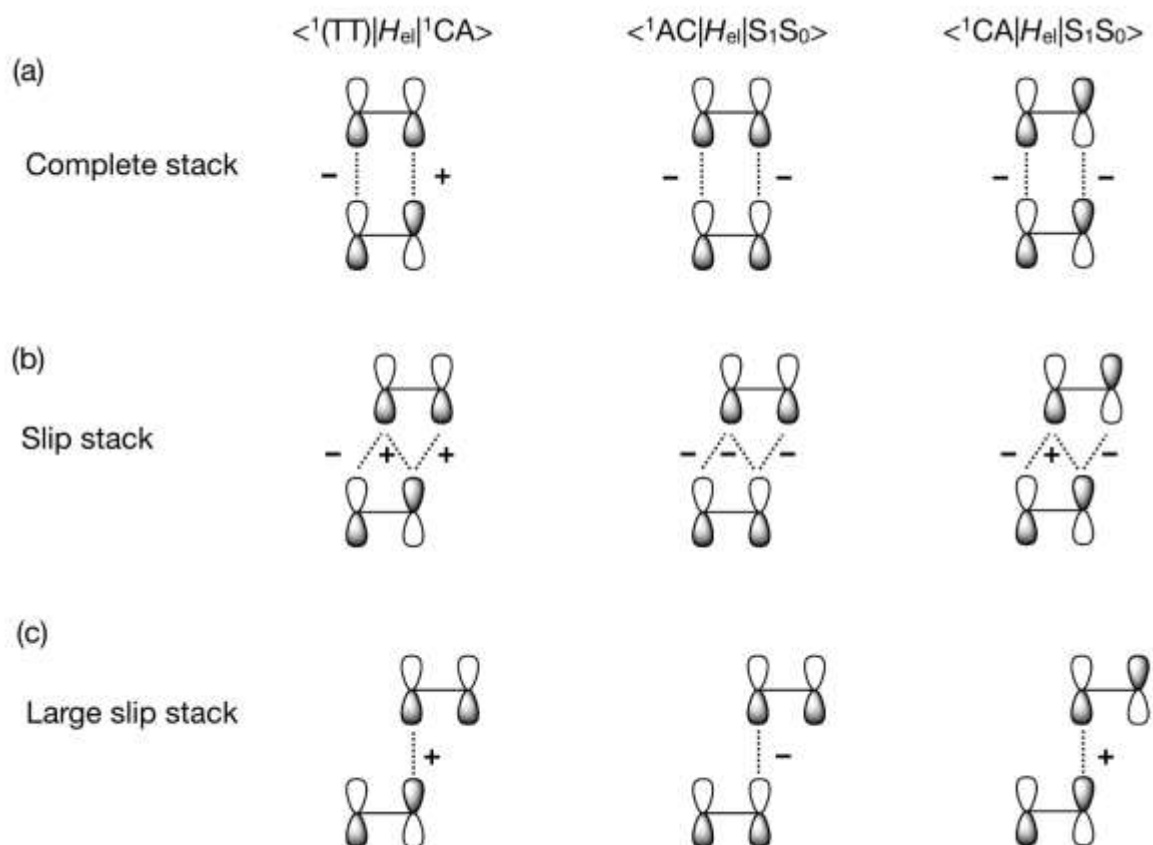


Fig. 13. Schematic diagram of orbital overlaps corresponding to Fock matrix elements (dominant) in Eqs. (9a–d) for complete-, slip-, and large-slip-stack ethylene dimers. The dotted lines indicate the interaction between upper and lower ethylene π -orbitals and the signs (+,–) indicate the positive and negative overlaps, respectively.

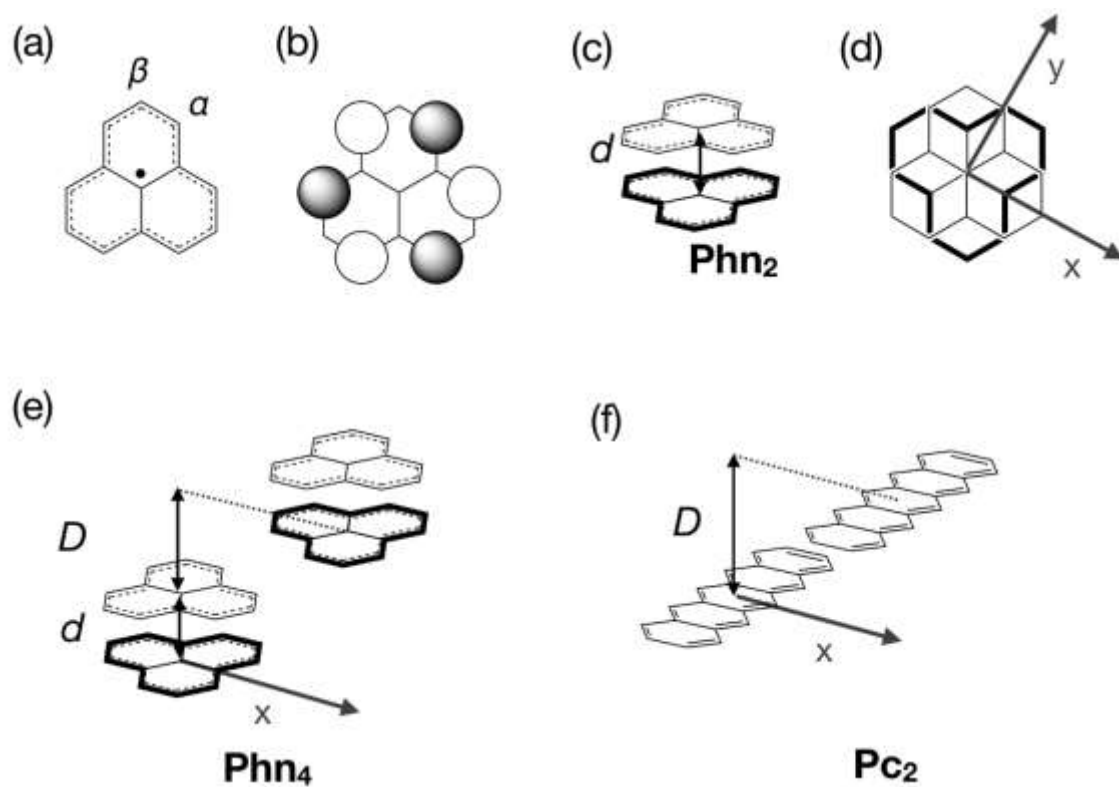


Fig. 14. Phenalenyl radical **Phn** (a), SOMO of **Phn** (b), staggered dimer **Phn₂** (c), top view of **Phn₂** (d), phenalenyl tetramer **Phn₄** (e), and pentacene dimer **Pc₂** (f). The slip-stack axis (x-axis) is also shown. Reproduced from Ref. [151] with permission from the PCCP Owner Societies.

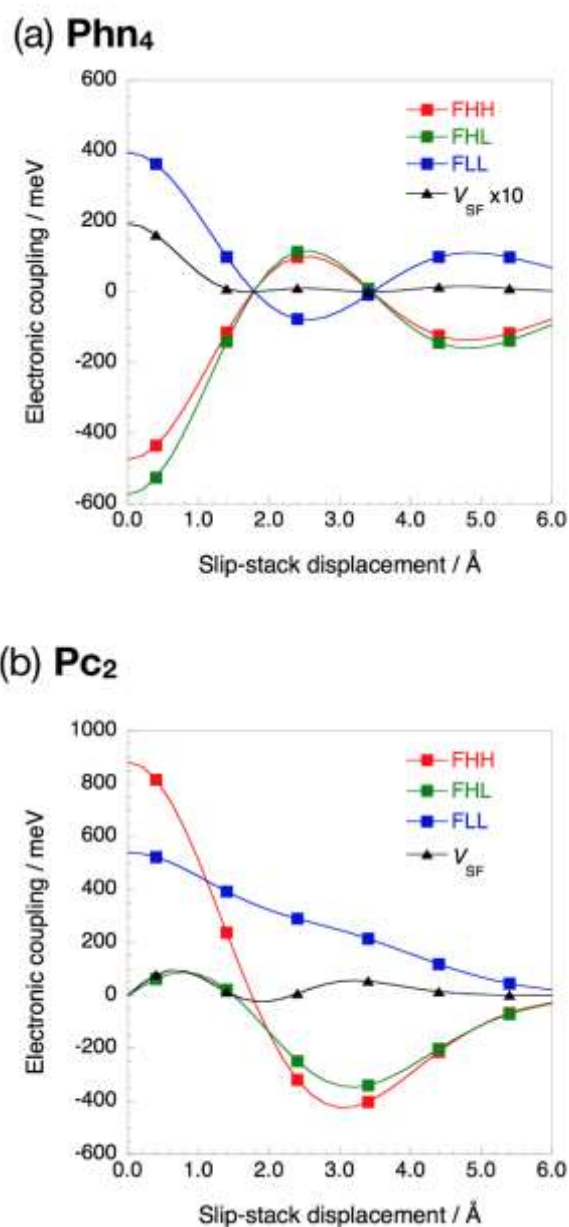


Fig. 15. Fock matrix elements, indicated by squares of red (F_{HH}), green (F_{HL}) and blue (F_{LL}), and SF coupling matrix element V_{SF} , indicated by black triangle, of **Phn₄** (a) and pentacene dimer (b). Intermonomer and interdimer distances, d and D , respectively, is set to 3.0 and 3.4 Å, respectively. Note that V_{SF} in **Phn₄** multiplied by 10 is shown. Reproduced from Ref. [151] with permission from the PCCP Owner Societies.

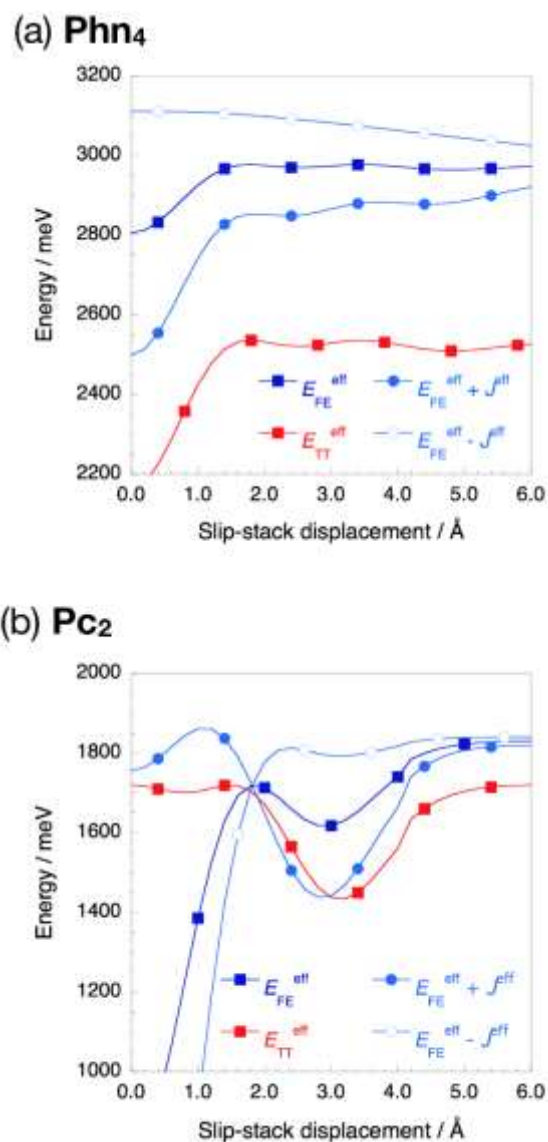


Fig. 16. Slip-stack displacement dependences of effective energies of FE' state E_{FE}^{eff} (blue filled square) and TT' state E_{TT}^{eff} (red filled square), and of the superpositions of FE' states split by exciton coupling $\pm J^{eff}$ (light blue filled circle for $E^{eff} + J^{eff}$ and light blue blank circle for $E^{eff} - J^{eff}$) in **Phn4** (a) and **Pc2** (b). Reproduced from Ref. [151] with permission from the PCCP Owner Societies.

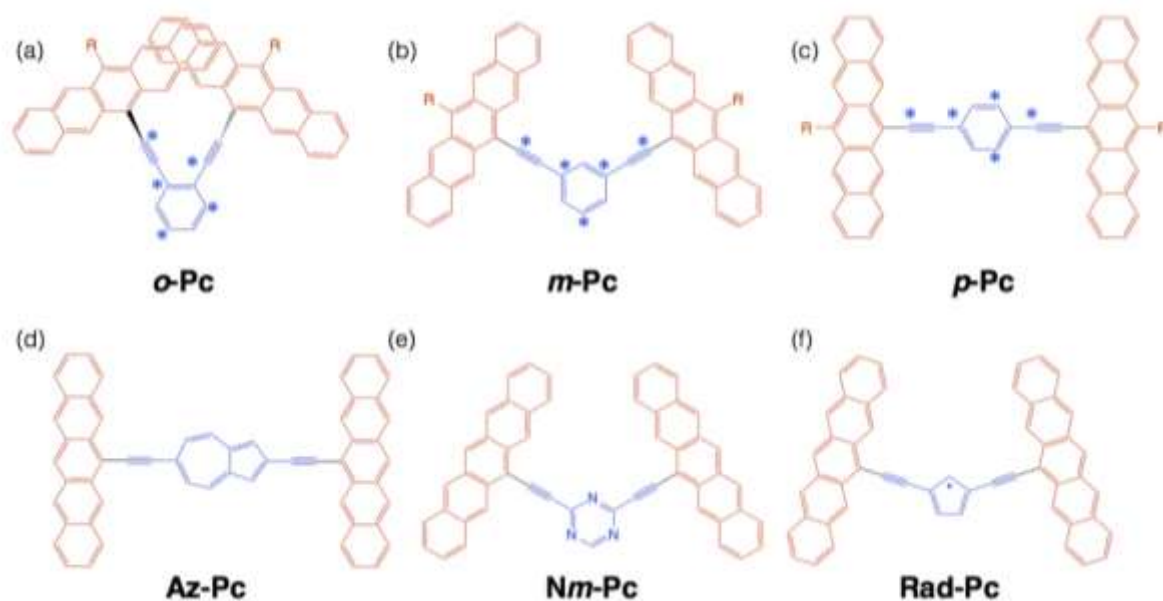


Fig. 17. Pentacene dimers linked with diethynylphenyl bridges, where the chromophores and bridge are indicated by red and blue lines, respectively. The substituent R in (a)-(c) is a hydrogen atom in the theoretical models. The carbons in the bridge of (a)-(c) are grouped into starred and non-starred carbons (see text). (d)-(f) are newly proposed molecules (see text). Reprinted with permission from Ref. [115]. Copyright © 2016 American Chemical Society.

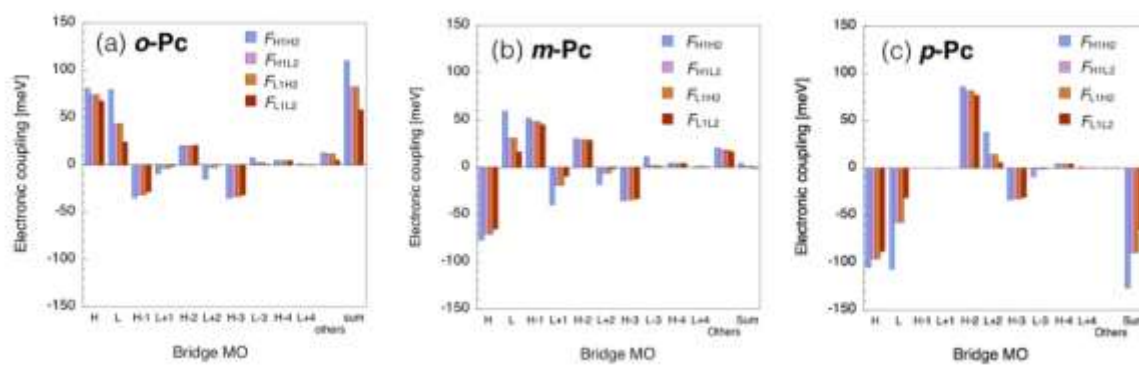


Fig. 18. Contributions through each bridge MO for the bridge-mediated coupling in (a) **o-Pc**, (b) **m-Pc** and (c) **p-Pc**. Here, the "B" of H and L indicating the bridge MO is omitted. "Sum" denotes the total sum of all the contributions. Reprinted with permission from Ref. [115]. Copyright @ 2016 American Chemical Society.

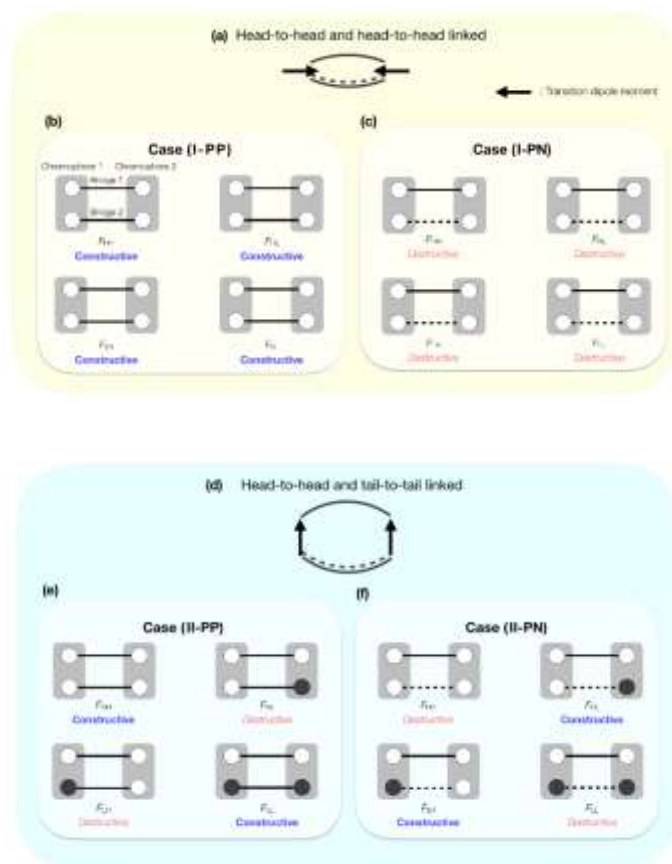
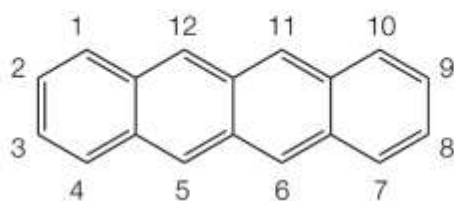


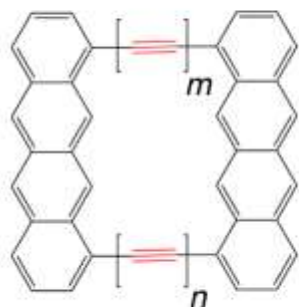
Fig. 19. Schematic diagrams of linked patterns of bridges with head-to-head and head-to-head positions (concerning the transition dipole moments of chromophores) (a), and with head-to-head and tail-to-tail positions (d). Quantum interference patterns, the cases (I-PP), (I-PN), (II-PP) and (II-PN) are shown in (b), (c), (e) and (f), respectively, where two chromophores (grey rectangular part) are linked by two bridges (black solid and/or broken lines). The phases of HOMO and LUMO at the linked sites are indicated by white (positive) and black (negative) circles. A positively-constructive and negatively-constructive bridges are described as black solid and broken lines, respectively. Ref. [116] - Published by The Royal Society of Chemistry.

(a)

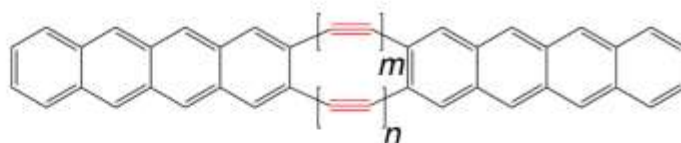


Tetracene

(b)



Tc1-(m,n)



Tc2-(m,n)

Fig. 20. Molecular structure of tetracene (a) and model doubly-bridged tetracene dimers (b). Here, $(m,n) = (1,1), (2,1), (2,2), (3,1), (3,2)$ and $(3,3)$. Bridge moieties are indicated by red lines. Ref. [116] -

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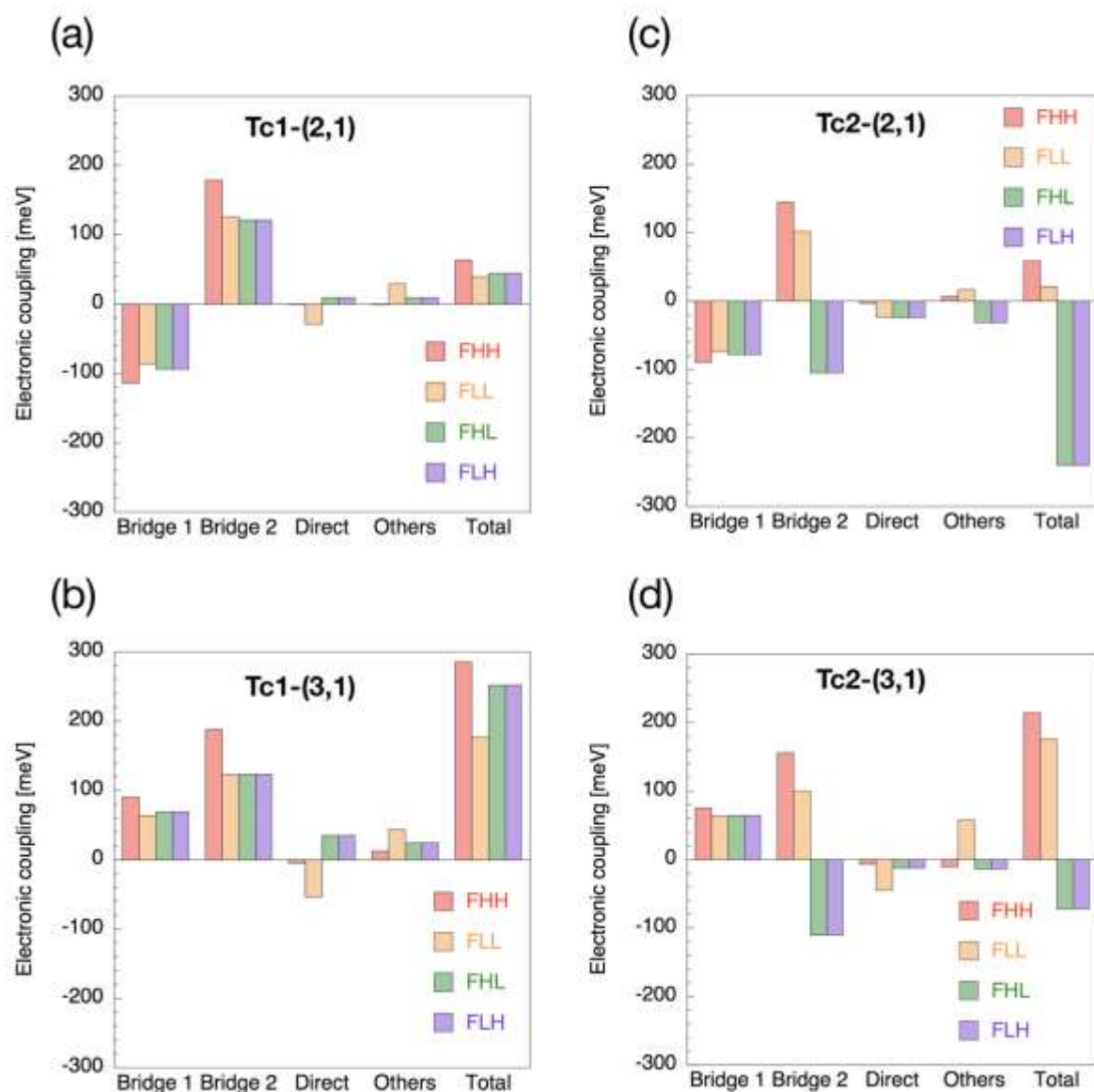


Fig. 21. Decomposition of the Fock matrix elements in **Tc1-(m,n)** and **Tc2-(m,n)** into the two bridge-mediated contributions from the bridges 1 and 2, direct-overlap contribution, and others. Here, $(m,n) = (2,1)$ ((a) and (c)) and $(3,1)$ ((b) and (d)). The longer bridge is referred to as bridge 1. Ref. [116] - Published by The Royal Society of Chemistry.

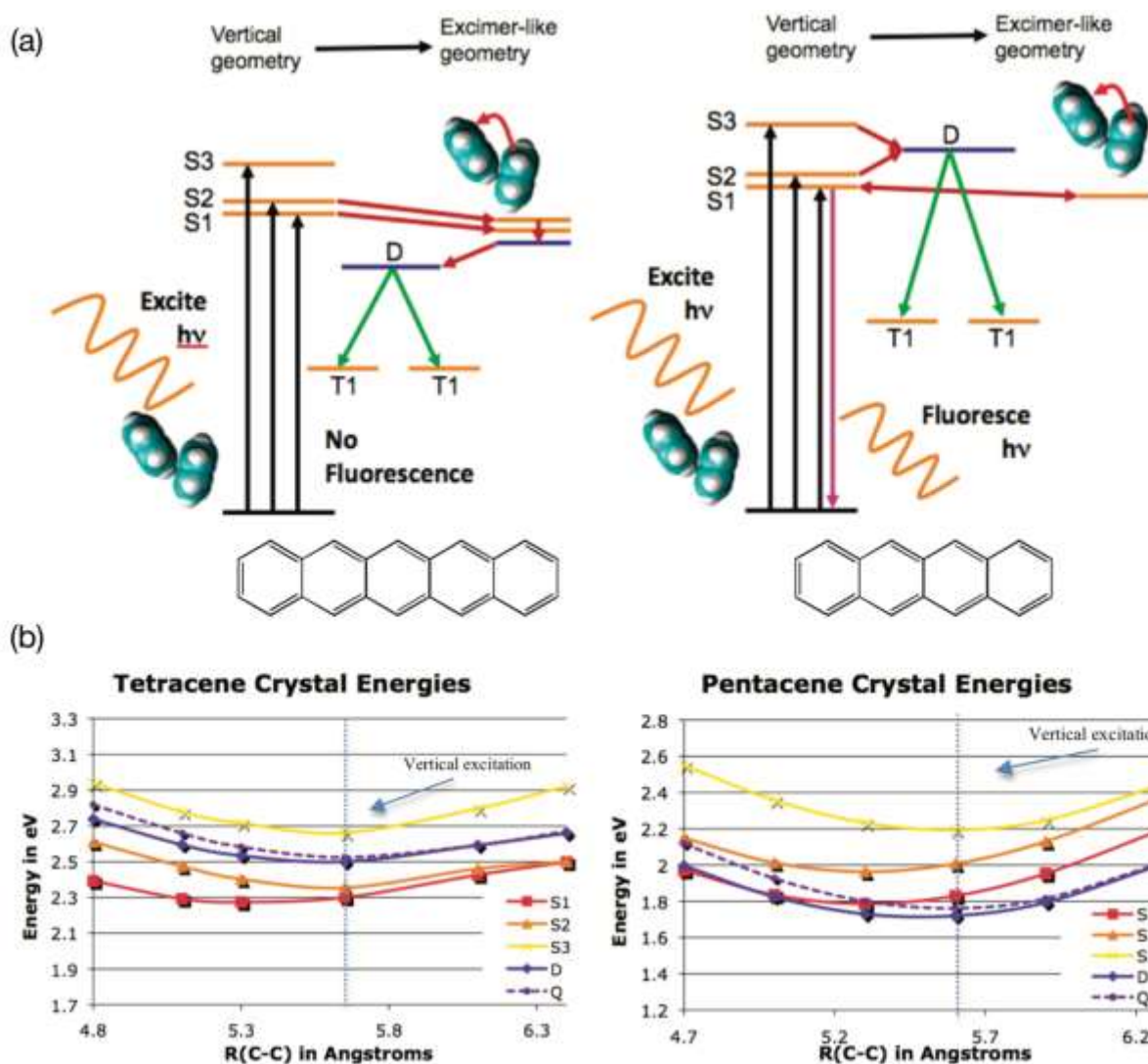


Fig. 22. (a) Singlet fission mechanism for pentacene and tetracene (b) potential energy surfaces along the $R(C-C)$ coordinate for tetracene and pentacene, including the lowest four singlet excited states (single-exciton S1–S3 and multi-exciton D) and the quintet (Q). Active dimer model is shown with constraint coordinate $R(C-C)$, which is the distance between the outermost carbon atoms on one side of each pentacene/tetracene monomer. Adapted with permission from Ref. [159].

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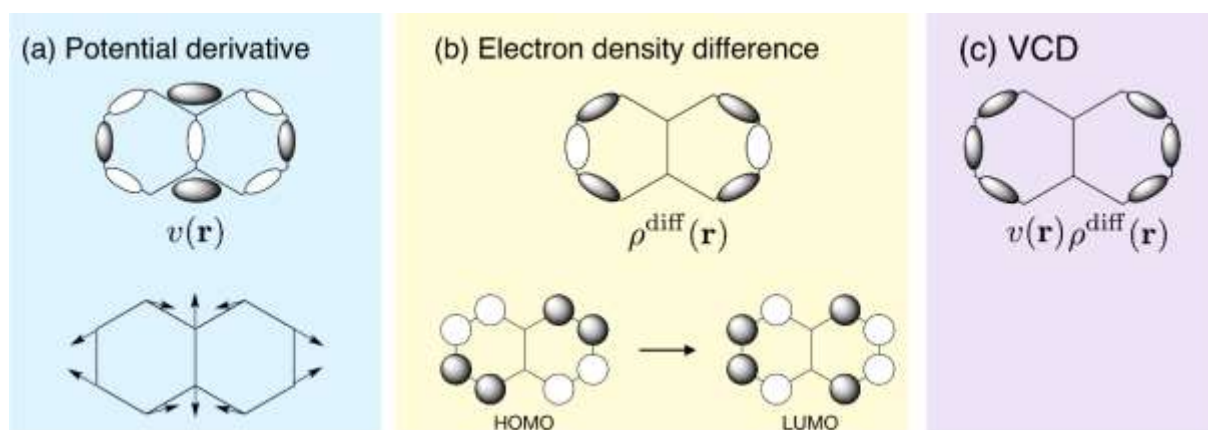


Fig. 23. (a) C-C stretching mode of naphthalene and its potential derivative $v(\mathbf{r})$, (b) electron density difference distribution $\rho^{\text{diff}}(\mathbf{r})$ as well as the HOMO and LUMO distributions, and (c) the VCD, which is obtained by the product of $v(\mathbf{r})$ and $\rho^{\text{diff}}(\mathbf{r})$, together with the corresponding nuclear motion.

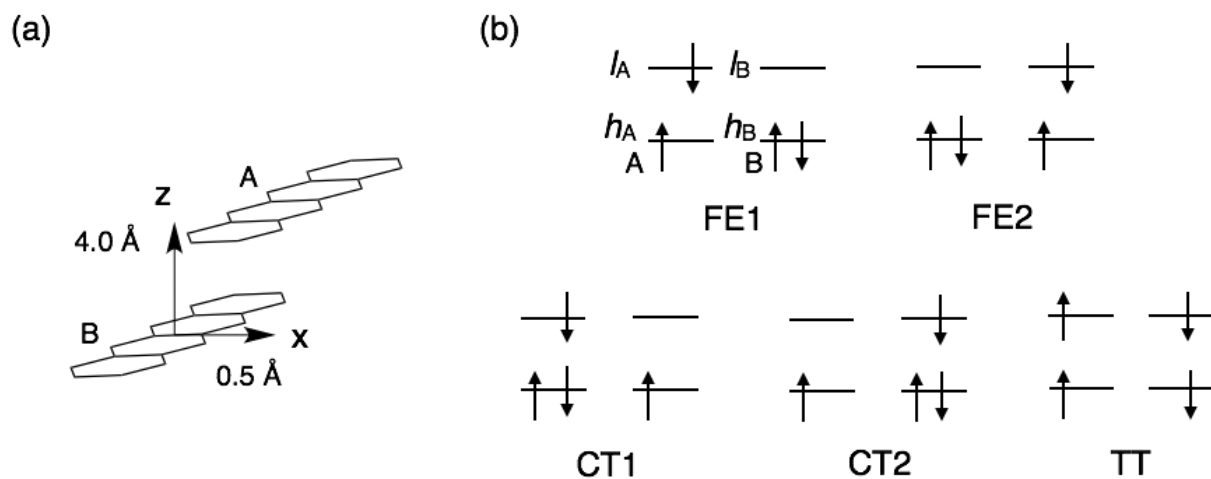


Fig. 24. (a) Tetracene dimer model and (b) five diabatic electronic states for singlet fission (see also Fig. 10). The (frozen) monomer orbitals, h_A , h_B , l_A and l_B , are calculated at restricted Hartree-Fock (RHF)/cc-pVDZ level of theory. Reprinted with permission from Ref. [162]. Copyright @ 2015 American Chemical Society.

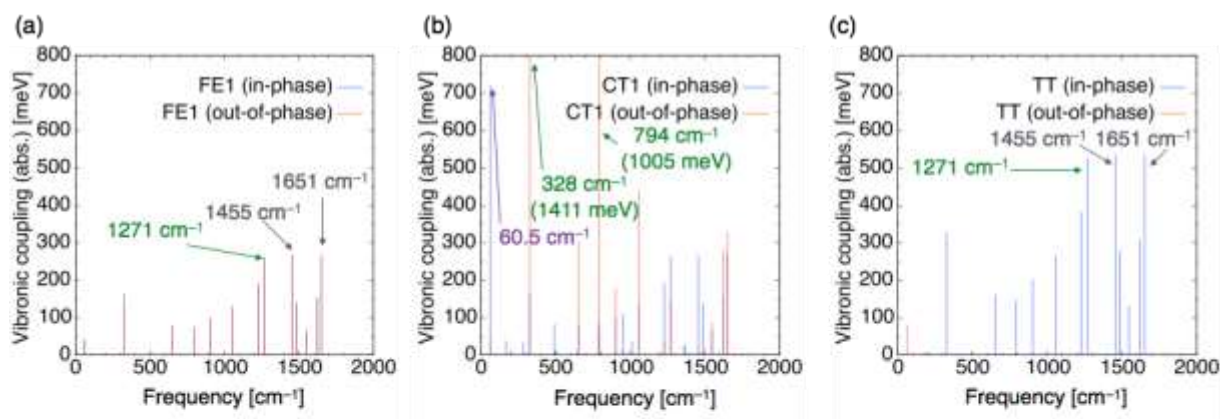


Fig. 25. Holstein couplings of FE1 (a), CT1 (b), and TT (c). The blue and red lines represent “in-phase” and “out-of-phase” vibration contributions, respectively. The purple, green, and grey numbers denote the frequencies (the values in parentheses indicate the vibronic coupling amplitudes) of out-of-plane bending modes, of the ring-breathing modes, and of the C-C stretching modes, respectively. Adapted with permission from Ref. [162]. Copyright @ 2015 American Chemical Society.

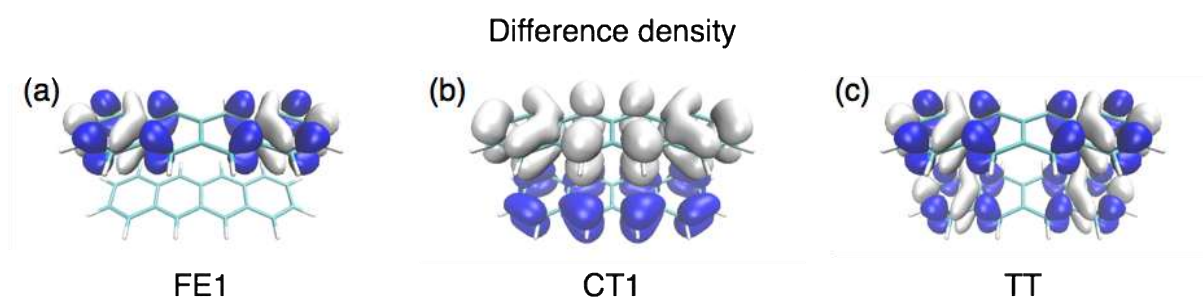


Fig. 26. Difference densities for FE1 (a), CT1 (b) and TT (c). The white and blue surfaces indicate positive and negative isosurfaces ($\pm 9 \times 10^{-4}$ a.u.), respectively. Adapted with permission from Ref. [162]. Copyright @ 2015 American Chemical Society.

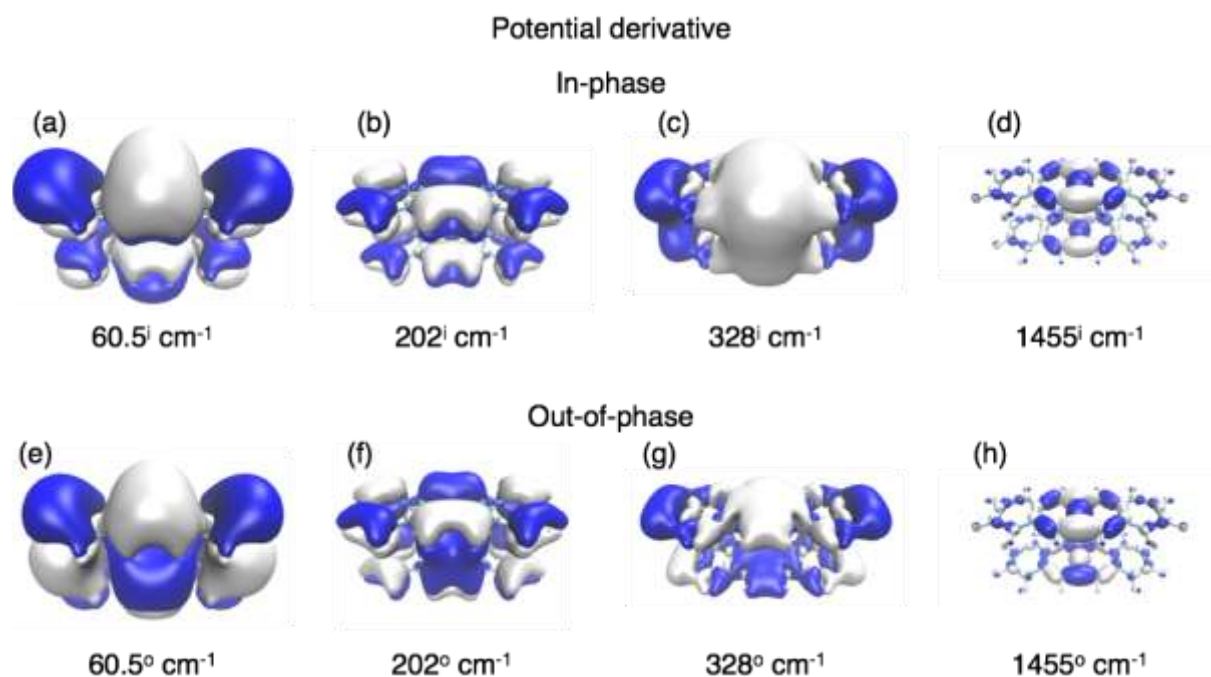


Fig. 27. Potential derivative maps of in-phase vibrational motions, 60.5 cm⁻¹ (a), 202 cm⁻¹ (b), 328 cm⁻¹ (c) and 1455 cm⁻¹ (d), as well as of out-of-phase vibrational motions, 60.5 cm⁻¹ (e), 202 cm⁻¹ (f), 328 cm⁻¹ (g) and 1455 cm⁻¹ (h). The white and blue surfaces represent the isosurfaces ± 0.05 a.u., respectively. Reprinted with permission from Ref. [162]. Copyright @ 2015 American Chemical Society.

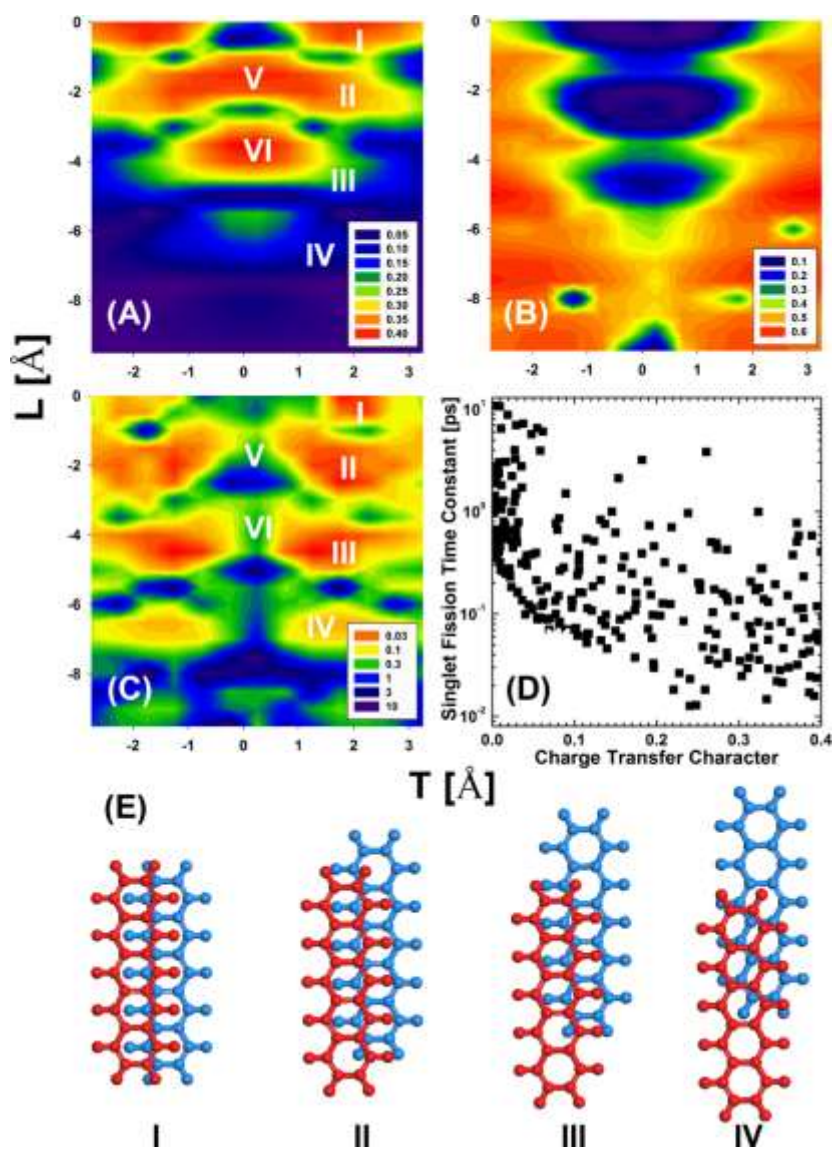


Fig. 28. 2D maps of the (A) CT character of the photoexcited state at time zero, (B) yield of the TT state at 50 ps, (C) SF time constant in ps, characteristic packings indicated with Roman numbers, (D) SF time constant as a function of the initial CT character, and (E) Intermolecular packings for the specified configurations. Reprinted with permission from Ref. [97]. Copyright @ 2014 American Chemical Society.

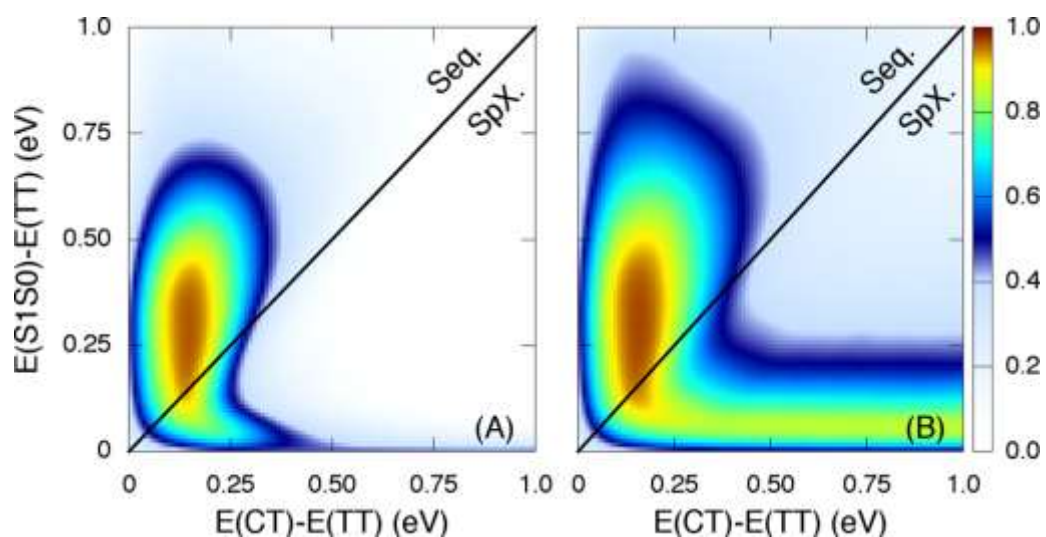


Fig. 29. Population of TT after 1 ps for $\lambda_p = 0$ (a) and 0.5 (b) meV. The other parameters are: $V_{lh} = V_{hl} = 15$ meV, $V_{2e} = 1$ meV, $\lambda_H = 50$ meV, $\Omega_H = \Omega_P = 150$ meV, and $V_{ll} = -V_{hh} = 25$ meV. Reprinted with permission from Ref. [123]. Copyright @ 2015 American Chemical Society.

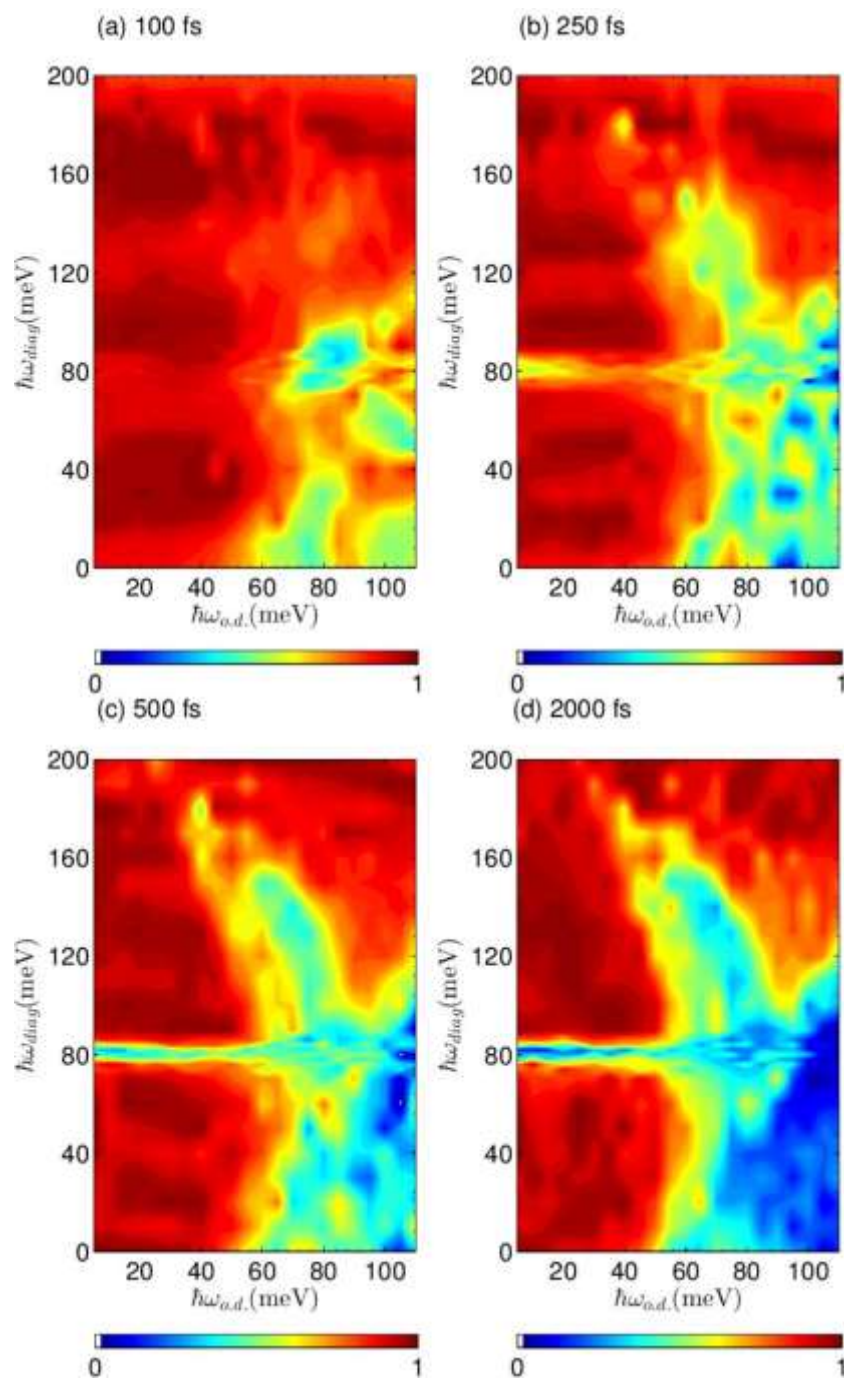


Fig. 30. Snapshots of singlet population as functions of $\hbar\omega_{diag}$ and $\hbar\omega_{o,d}$ at the time of (a) 100 fs, (b) 250 fs, (c) 500 fs, and (d) 2000 fs. The other parameters are $J_{S1,TT} = 20$ meV and $\epsilon_{S1,g} - \epsilon_{TT,g} = 100$ meV.

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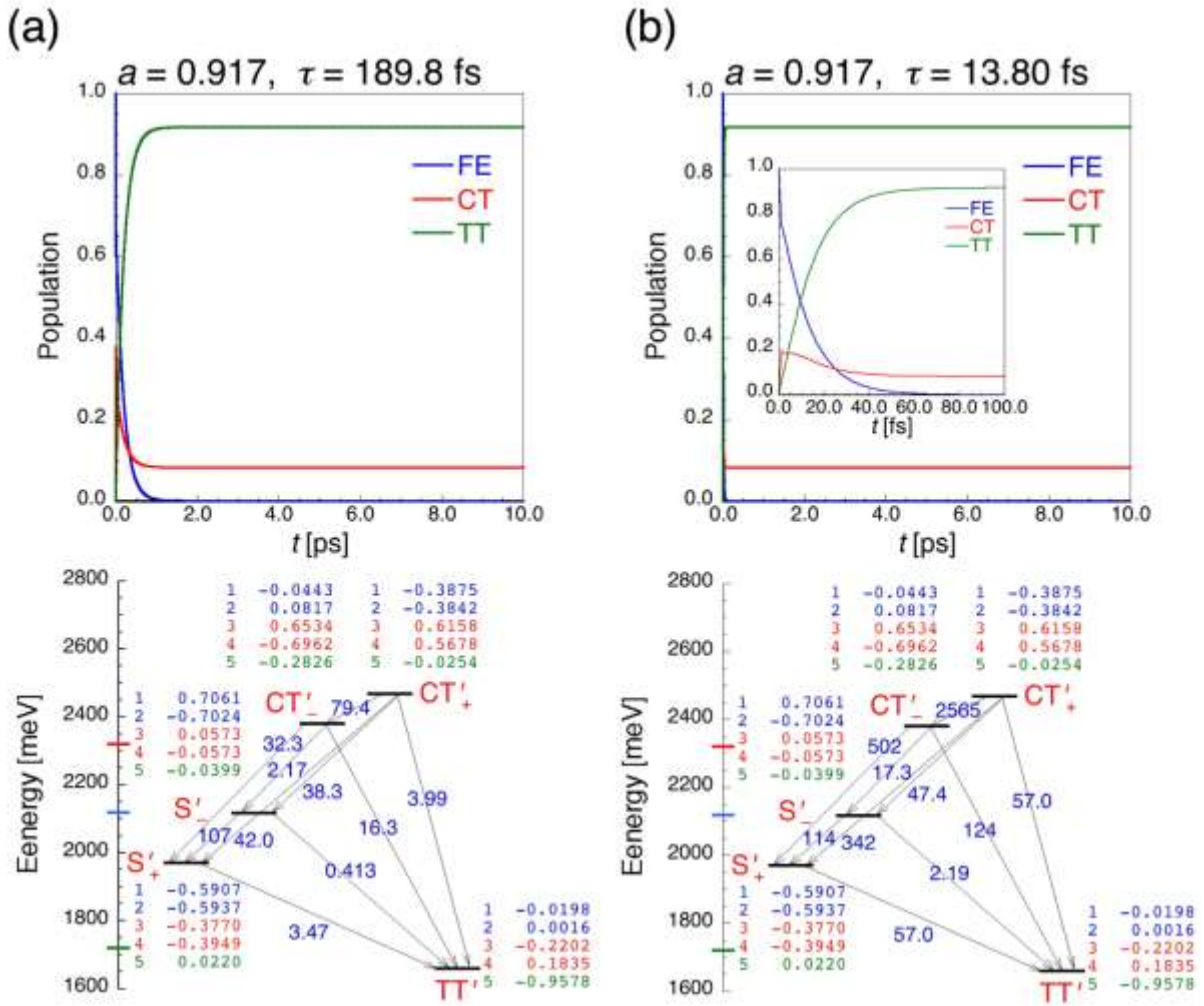


Fig. 31. Time evolution of diabatic exciton populations $\{P(\text{FE}), P(\text{CT}), P(\text{TT})\}$ and adiabatic state energy diagram of $\{|S_0\rangle, |S_0'\rangle, |CT_1\rangle, |CT_2\rangle, |TT\rangle\}$ for the pentacene dimer model with identical spectral density with $(\Gamma, W) = (50, 180)$ meV for each diabatic state (a), and that with state-dependent spectral densities $((\Gamma_m, W_m) = (50, 180)$ meV for FE, (1290, 98) meV for CT, and (200, 180) meV for TT (b). For (b), the time evolution in the early time region is shown in the inset. In the adiabatic state energy diagram, the coefficients C_{ma} ($\{|m\rangle\} = \{|1\rangle, |2\rangle, |3\rangle, |4\rangle, |5\rangle\} = \{|S_0S_0\rangle, |S_0S_1\rangle, |CA\rangle, |AC\rangle, |TT\rangle\}$) for each adiabatic state $\{|a\rangle\} = \{|S_0\rangle, |S_0'\rangle, |CT_1\rangle, |CT_2\rangle, |TT\rangle\}$ are shown, and the direction of relaxation is shown by an arrow with a relative relaxation factor [meV]. The diabatic state energies, $E(\text{FE})$, $E(\text{CT})$ and $E(\text{TT})$, are indicated by blue, red, and green bars on the energy axis, respectively. Reprinted with permission from Ref. [168]. Copyright @ 2016 American Chemical Society.

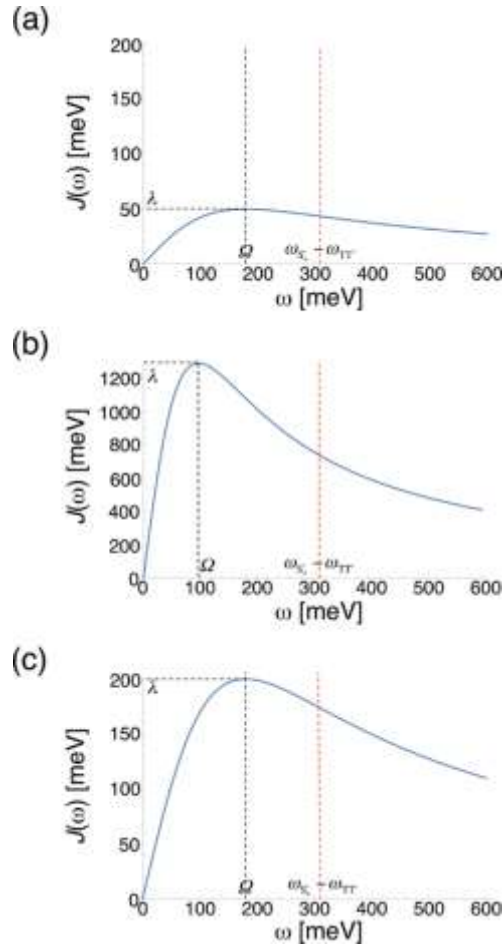


Fig. 32. Spectral densities (Eq. (45c)) for three cases: (a) $(\lambda, \Omega) = (50, 180)$ meV, (b) $(1290, 98)$ meV, and (c) $(200, 180)$ meV. The vertical black and red dashed lines indicate the cutoff frequency Ω and $\omega_{S\phi} - \omega_{TT\phi} = 310.7$ meV for the pentacene dimer model, respectively. Reprinted with permission from Ref. [168]. Copyright @ 2016 American Chemical Society.

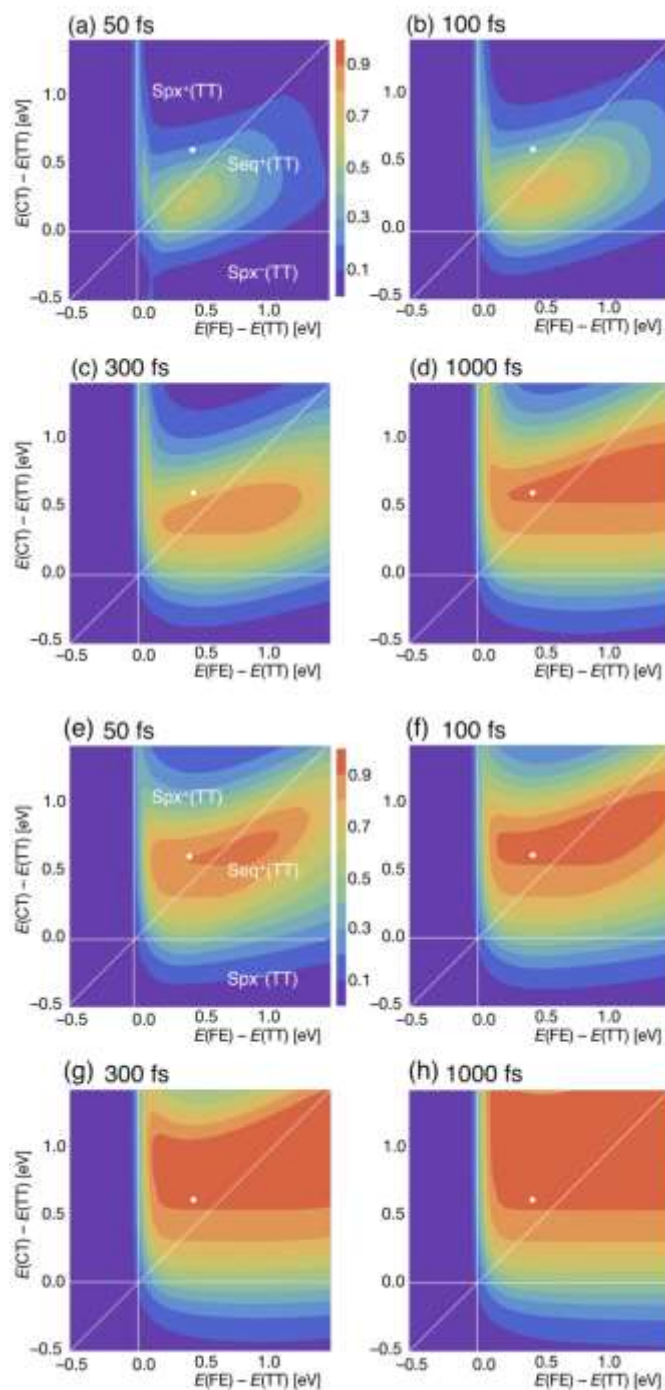


Fig. 33. Double-triplet yield $P(TT)$ contour map on the $(E(FE) - E(TT))-(E(CT) - E(TT))$ plane after 50 fs, 100 fs, 300 fs and 1000 fs for the realistic pentacene dimer model with identical spectral density ($(\Gamma_m, W_m) = (50, 180)$ meV for each state) (a-d) and with state-dependent spectral densities ($(\Gamma_m, W_m) = (50, 180)$ meV for FE, (1290, 98) meV for CT, and (200, 180) meV for TT) (e-h). The white dot indicates the energy region for the realistic pentacene dimer. Adapted with permission from Ref. [168]. Copyright @ 2016 American Chemical Society.

